

DIFFRACTION AND SCATTERING OF IONIZING RADIATIONS

Dedicated to the 60th Birthday of M.V. Kovalchuk

Neutron Standing Waves in Layered Systems

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Received March 27, 2006

Abstract—A theory of propagation of neutron waves in multilayer magnetic and nonmagnetic systems is presented. It is shown how a system forms the wave function of a neutron, how the wave function formed can be controlled, and how this function can be used to investigate materials. The results of experimental investigations of the neutron reflection from multilayer systems with simultaneous measurement of different types of secondary radiation are reported.

PACS numbers: 61.12.–q

DOI: 10.1134/S1063774506050038

1. INTRODUCTION

During diffraction of X rays or neutrons from periodic structures, a standing wave field is formed as a result of the interference of the wave fields of the incident and scattered particles [1–3]. In ideal crystals, the planes of equal intensity in the wave field are located parallel to the diffraction reflection planes and exist both in the bulk and above the surface. This circumstance makes it possible to determine with high accuracy the position of impurity atoms in both crystals and layers adsorbed on the crystal surface. The standing wave period is close to the lattice parameter and lies in the range 0.1–10 Å. Large periods (about several hundred angstroms) can be attained with the use of multilayer synthetic microstructures with a specified period for standing-wave generation [4].

A standing-wave field also arises upon total reflection from mirrors [5]. In this case, the standing-wave period is $D \propto \lambda/\theta$, where θ is the grazing angle of a particle beam. This relation makes it possible to change the value of D from several hundred angstroms near the critical angle to several thousand angstroms at smaller angles and opens new possibilities of investigating long-period structures (in particular, biological objects).

The above-mentioned possibilities of standing waves were successfully implemented in X-ray physics. A large contribution to the development of this new method for studying the structures of crystals and surfaces was made by the researchers from the Institute of Crystallography of the Russian Academy of Sciences

under the guidance of Corresponding Member of the Russian Academy of Sciences M.V. Koval'chuk [6].

The intensity of neutron standing waves had remained low until recently and, therefore, their practical application was limited. After the first observation of neutron standing waves in 1956 [1], several interesting studies which demonstrated the fundamental possibilities of neutron standing waves in crystals have been carried out [7–12].

The situation changed when it was shown that the amplitude of standing waves in thin films can be resonantly amplified for both X rays [13, 14] and neutrons [15, 16] in the case of total reflection. It was shown that, under certain conditions, the gain of neutron standing waves can be as high as several hundred.

In [15, 16], a conventional method was used to detect γ photons (secondary radiation) that arise as a result of intranuclear processes after trapping of a neutron. A sample was prepared in such a way as to supply the layer under study with atomic nuclei having large cross sections of the (n, γ) reaction. For example, in [16], a gadolinium-containing layer was embedded in a polymer film. Other types of secondary radiation, for example, α particles [17], may also be present in reactions with neutrons.

It was shown for the first time in [18, 19] that, under conditions of total reflection of polarized thermal neutrons, a neutron standing wave is formed in a specified spin state (with the spin component oriented approximately along the magnetic field or in the opposite direc-

tion). This wave can be recorded by detecting spin-flip reflected neutrons.

The experiments that have been performed in recent years initiated the development of theoretical problems of a general character that concern the propagation of neutron waves in cavity systems [20, 21].

Thus, the method of neutron standing waves has been fundamentally developed in the last decade. It is especially important in view of the improvement of nanotechnologies, which make it possible to prepare layered structures with variation in the parameters and properties of individual layers. Therefore, we believe it is timely to describe the theoretical fundamentals of this method and the corresponding experimental data.

The problem can be formulated as follows. Upon formation of a standing wave, the wave function of a neutron acquires maxima and minima that are fixed in space; i.e., a certain structure of the wave function is formed. It is necessary to calculate this structure and reveal it with particular probes.

The formation of a structure of wave functions is most pronounced in resonant systems, i.e., systems with the optical potential shown in Fig. 1, where a layer with a number i is located between other layers. The left and right parts of this system are denoted as 1 and 2, respectively.

Resonant systems make it possible to (i) study weakly perturbed structures, in which the probability of interactions with neutrons is extremely low and does not exceed 10^{-10} [18]; (ii) form sources of coherent and polarized neutron radiation [22, 23] with a divergence as low as 10^{-9} rad and a monochromaticity as high as $\partial\lambda/\lambda = 10^{-7}$; (iii) develop new polarization neutron techniques in which the phase difference between the waves of two spin states of a neutron in a magnetic field is measured, as a result of which the spectrum of low-frequency excitations or density correlation lengths in a medium can be recorded through measurement of ultras-small transfers (down to 10^{-6} Å $^{-1}$) of the neutron wave vector [24]; and (iv) study the coherent properties of neutron radiation.

In Sections 2–5, the theory of propagation of neutrons in multilayer magnetic and nonmagnetic systems is described and, in Section 6, a review of the experiments performed on the IBR-2 reactor in Dubna is given.

2. NEUTRONS IN A MULTILAYER STRUCTURE

All problems considered in this study are related to neutron optics, where the reflectance and transmittance of plane multilayer mirrors and the wave function in a mirror are investigated. A plane mirror is represented by some potential [25] and the scattering of a neutron from the mirror is found by solving the 1D stationary Schrödinger equation with a specified incident plane wave and the mirror potential [26].

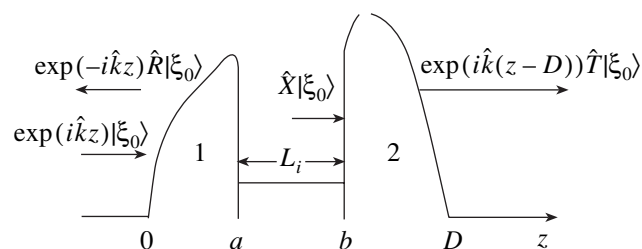


Fig. 1. Derivation of the neutron wave function in a layer i .

Let us consider the general problem of reflectance and transmittance of an arbitrary multilayer system with some isolated homogeneous layer with a number i . All other parts of the system will be denoted by abstract potentials 1 and 2, as shown in Fig. 1. For generality, we will take into account the neutron spin and assume the magnetic field to be uniform. Then, we will simplify the problem and, after switching off the magnetic field, disregard the spin properties of the neutron.

2.1. A Magnetic Multilayer System

Let a neutron be incident on a magnetic system located in an external magnetic field $\mathbf{B}_0$ and represented by the potential shown in Fig. 1 from the left. The neutron, described by the plane wave $\exp(ikz)|\xi_0\rangle$, is in the spin state $|\xi_0\rangle$, where $\hat{k} = \sqrt{k_0^2 - 2\sigma\mathbf{B}_0}$ is the matrix of the normal-to-mirror component of the neutron wave vector in the field $\mathbf{B}_0$, k_0 is the normal-to-mirror component of the neutron wave vector in vacuum, $\sigma = (\sigma_x, \sigma_y, \sigma_z)$ is the Pauli matrix, and z is a coordinate along the normal to the multilayer system. The layers are assumed to be infinitely extended and homogeneous along the interface; therefore, the problem is 1D. The external magnetic field $\mathbf{B}_0$ on the left and right from the mirror will be assumed identical.

Let us place the origin of the coordinate ($z = 0$) at the input surface of the first layer of the system; the incident wave at this point is in the state $|\xi_0\rangle$. Let us denote the wave incident on the part of the potential on the right from the layer i under study and within this layer as $\exp(ik_i(z - b))\hat{X}|\xi_0\rangle$ and the waves reflected from the entire system and transmitted through it as $\exp(-ikz)\hat{R}|\xi_0\rangle$ and $\exp(ik(z - D))\hat{T}|\xi_0\rangle$, respectively. Here, D is the total thickness of the mirror; b is the coordinate of the right edge of the layer i ; $\hat{X}$, $\hat{R}$, and $\hat{T}$ are 2×2 matrices; and $\hat{k}_i = \sqrt{k_0^2 - u_i - 2\sigma\mathbf{B}_i}$ is the matrix normal-to-mirror component of the wave vector in the layer i with the nuclear potential u_i and the magnetic field $\mathbf{B}_i$. Below, for simplicity, we will refer to the normal-to-mirror component of the wave vector as the wave vector.

It is easy to construct the following equations for the matrices $\hat{X}$, $\hat{R}$ and $\hat{T}$:

$$\hat{X} = \exp(i\hat{k}_i L_i) \hat{T}_1 + \exp(i\hat{k}_i L_i) \hat{R}_1 \exp(i\hat{k}_i L_i) \hat{R}_2 \hat{X}, \quad (1)$$

$$\hat{R} = \hat{R}_1 + \hat{T}_1 \exp(i\hat{k}_i L_i) \hat{R}_2 \hat{X}, \quad \hat{T} = \hat{T}_2 \hat{X}. \quad (2)$$

Here, $\hat{R}_i$ ($i = 1, 2$) is the matrix of amplitudes of reflection from potentials 1 and 2 on the left, $\hat{R}_i$ is the matrix of amplitudes of reflection from these potentials on the right, $\hat{T}_i$ is the matrix of amplitudes of transmission of potentials 1 and 2 from left to right, $\hat{T}_i$ is the matrix of amplitudes of transmission of potentials 1 and 2 from right to left, and L_i is the width of the layer i .

Equation (1) shows that a wave with the amplitude $\hat{X}$ is equal to the sum of two waves, one of which is the part of incident wave transmitted through potential 1 and the other is generated by the wave with the amplitude $\hat{X}$ that is incident on potential 2. Equation (2) shows that the reflected wave is also equal to the sum of two waves, one of which is the part of the incident wave that is reflected only from potential 1, while the other is generated by the wave with the amplitude $\hat{X}$ that is incident on potential 2 and, after reflection from potential 2, returns to external space through potential 1.

The transmitted wave is proportional to $\hat{X}$ and the transmission amplitude of potential 2.

Solution of system (1), (2) is elementary:

$$\hat{X} = [\hat{I} - \exp(i\hat{k}_i L_i) \hat{R}_1 \exp(i\hat{k}_i L_i) \hat{R}_2]^{-1} \times \exp(i\hat{k}_i L_i) \hat{T}_1, \quad (3)$$

$$\begin{aligned} \hat{R} &= \hat{R}_1 \\ &+ \hat{T}_1 \exp(i\hat{k}_i L_i) \hat{R}_2 [\hat{I} - \exp(i\hat{k}_i L_i) \hat{R}_1 \exp(i\hat{k}_i L_i) \hat{R}_2]^{-1} \\ &\times \exp(i\hat{k}_i L_i) \hat{T}_1, \end{aligned} \quad (4)$$

$$\begin{aligned} \hat{T} &= \hat{T}_2 \hat{X} \\ &= \hat{T}_2 [\hat{I} - \exp(i\hat{k}_i L_i) \hat{R}_1 \exp(i\hat{k}_i L_i) \hat{R}_2]^{-1} \exp(i\hat{k}_i L_i) \hat{T}_1, \end{aligned} \quad (5)$$

where $\hat{I}$ is a unit matrix. When $\hat{X}$ is known, it is easy

to find the wave function $|\psi_i\rangle$ in the layer i :

$$\begin{aligned} |\psi_i\rangle &= [\exp(i\hat{k}_i(z-b)) + \exp(-i\hat{k}_i(z-b)) \hat{R}_2] \hat{X} |\xi_0\rangle \\ &= [\exp(i\hat{k}_i(z-b)) + \exp(-i\hat{k}_i(z-b)) \hat{R}_2] \\ &\times [\hat{I} - \exp(i\hat{k}_i L_i) \hat{R}_1 \exp(i\hat{k}_i L_i) \hat{R}_2]^{-1} \exp(i\hat{k}_i L_i) \hat{T}_1 |\xi_0\rangle. \end{aligned} \quad (6)$$

The matrices $\hat{R}$ and $\hat{T}$ of the reflection and transmission amplitudes contain diagonal elements R_{++} , R_{--} , T_{++} , and T_{--} . They describe the non-spin-flip reflection ($|R_{++}|^2$ and $|R_{--}|^2$) and transmission ($|T_{++}|^2$ and $|T_{--}|^2$) probabilities. In addition, the matrices $\hat{R}$ and $\hat{T}$ contain the off-diagonal elements R_{+-} , R_{-+} , T_{+-} , and T_{-+} , which describe the spin-flip reflection ($|R_{+-}|^2 k_-/k_+$ and $|R_{-+}|^2 k_+/k_-$) and transmission ($|T_{+-}|^2 k_-/k_+$ and $|T_{-+}|^2 k_+/k_-$) probabilities. Here, the subscripts $+$ and $-$ denote the polarizations that are parallel and antiparallel to the external field direction, respectively. The left subscript is associated with the incident neutron and the right subscript is associated with the reflected or transmitted neutrons. The ratio of the wave vectors $k_{\pm} = \sqrt{k_0^2 \mp 2B_0}$, where k_0 is the wave vector in a zero-field vacuum, characterizes the change in the neutron flux from the surface and is necessary to satisfy the unitarity condition.

2.2. A Nonmagnetic Multilayer System

If all fields in the entire space are zero, we have a solution for a zero-spin (i.e., a scalar) particle. In this case, all quantities become scalars, the expressions are simplified, and we obtain

$$X = \frac{\vec{T}_1 \exp(ik_i L_i)}{1 - \exp(2ik_i L_i) \vec{R}_1 \vec{R}_2}, \quad k_i = \sqrt{k_0^2 - u_i}, \quad (7)$$

$$\begin{aligned} R &= \vec{R}_1 + \vec{T}_1 \vec{T}_1 \frac{\exp(2ik_i L_i) \vec{R}_2}{1 - \exp(2ik_i L_i) \vec{R}_1 \vec{R}_2} \\ &= \frac{\vec{R}_1 + (\vec{T}_1 \vec{T}_1 - \vec{R}_1 \vec{R}_1) \exp(2ik_i L_i) \vec{R}_2}{1 - \exp(2ik_i L_i) \vec{R}_1 \vec{R}_2}. \end{aligned} \quad (8)$$

$$T = \frac{\vec{T}_2 \vec{T}_1 \exp(ik_i L_i)}{1 - \exp(2ik_i L_i) \vec{R}_1 \vec{R}_2}, \quad (9)$$

$$\begin{aligned} \psi_i &= [\exp(ik_i(z-b)) + \exp(-ik_i(z-b)) \vec{R}_2] \\ &\times \frac{\exp(ik_i L_i) \vec{T}_1}{1 - \exp(2ik_i L_i) \vec{R}_1 \vec{R}_2}. \end{aligned} \quad (10)$$

When absorption in the system can be disregarded,

the density of the scalar wave function in the layer i has the form

$$|\psi_i|^2 = \left(1 + |\vec{R}_2|^2 + 2|\vec{R}_2| \cos(2k_i(z-b) - \vec{\phi}_2) \right) \times \left| \frac{\vec{T}_1}{1 - \exp(2ik_i L_i) \vec{R}_1 \vec{R}_2} \right|^2, \quad (11)$$

where $\vec{\phi}_2 = \arg(\vec{R}_2)$.

If $2k_i L_i > 2\pi n$, the density of the wave function in a layer i has a minimum $\min(|\psi_i|^2) \propto (1 - |\vec{R}_2|^2)$ at the points $z_{\min} = b - [\pi(2m-1) - \vec{\phi}_2]/2k_i$ and a maximum $\max(|\psi_i|^2) \propto (1 + |\vec{R}_2|^2)$ at the points $z_{\max} = b - (2\pi m - \vec{\phi}_2)/2k_i$, where m and n are integers ($1 \leq m \leq n$). The ratio of the maximum to minimum density

$$\chi = \left(\frac{1 + |\vec{R}_2|^2}{1 - |\vec{R}_2|^2} \right)^2 \quad (12)$$

will be referred to as the contrast. The maximum value of expression (11) and the factor on the right will be referred to as the gain K and the resonant factor B , respectively. Substituting $\vec{R}_1 = |\vec{R}_1| \exp(i\vec{\phi}_1)$ and $\vec{R}_2 = |\vec{R}_2| \exp(i\vec{\phi}_2)$ into (11), we obtain

$$K = (1 + |\vec{R}_2|^2)^2 |B|^2 = \frac{|\vec{T}_1|^2 (1 + |\vec{R}_2|^2)^2}{|1 - \exp(i\xi(E)) \vec{R}_1 \vec{R}_2|^2},$$

where the phase $\xi(E)$ depends on the energy $E = k_0^2/2$ of the neutron motion along the normal to the system:

$$\xi(E) = 2k_i L_i + \vec{\phi}_1 + \vec{\phi}_2. \quad (13)$$

In the case of a resonance, we have

$$\xi(E) = 2\pi n, \quad (14)$$

where n is an integer, and the gain becomes

$$K = \frac{|\vec{T}_1|^2 (1 + |\vec{R}_2|^2)^2}{(1 - |\vec{R}_1| |\vec{R}_2|)^2}. \quad (15)$$

Note that, since the potential u_i is nonzero, in view of the flux continuity condition, the following equalities are valid when losses can be neglected:

$$\begin{aligned} |\vec{R}_1|^2 &= 1 - \frac{k_i}{k_0} |\vec{T}_1|^2, & |\vec{R}_1|^2 &= 1 - \frac{k_0}{k_i} |\vec{T}_1|^2, \\ |\vec{R}_2|^2 &= 1 - \frac{k_0}{k_i} |\vec{T}_2|^2. \end{aligned} \quad (16)$$

If $|\vec{R}_1|$ and $|\vec{R}_2|$ are close to unity (i.e., $|\vec{T}_1| \ll 1$ and $|\vec{T}_2| \ll 1$) and the condition $2k_i L_i + \vec{\phi}_1 + \vec{\phi}_2 = 2\pi n$ is satisfied, the gain is reduced to the form

$$K \approx \frac{16k_i^2 |\vec{T}_1|^2}{k_0^2 (|\vec{T}_2|^2 + |\vec{T}_1|^2)^2}. \quad (17)$$

It follows from (17) that the value of K may greatly exceed unity.

Thus, fitting potentials 1 and 2 and the layer length L_i , one can significantly increase the wave function amplitude in the layer i [20] and the sensitivity of the experiments in which the wave function amplitude is most important [2].

Note that expressions (7)–(11) are valid at real and complex potentials 1 and 2 and u_i , i.e., when there are losses in the material [25]. If the potential u_i contains an imaginary part, the resonance condition takes the form

$$\xi(E) = 2k_i' L_i + \vec{\phi}_1 + \vec{\phi}_2 = 2\pi n, \quad (18)$$

where k_i' is the real part of the wave vector in the layer i . In this case, the gain is

$$K = \frac{|\vec{T}_1|^2 (1 + |\vec{R}_2|^2)^2}{(1 - \exp(-2k_i'' L_i) |\vec{R}_1| |\vec{R}_2|)^2}, \quad (19)$$

where k_i'' is the imaginary part of the wave neutron vector in the layer i .

2.3. The Width of Resonances in Resonant Systems

Expressions (7)–(10) contain the resonant factor

$$B(E) = \exp(-i\xi(E)) \frac{\vec{T}_1}{\exp(-i\xi(E)) - \exp(-2k_i'' L_i) \vec{R}_1 \vec{R}_2}, \quad (20)$$

where the phase $\xi(E)$ is defined in (13) and the exponential $\exp(-2k_i'' L_i)$, which characterizes absorption in the film, is explicitly singled out in the denominator. Let us show that, near the resonance (18), expression (20) can be reduced to the Breit–Wigner formula

$$B(E_n) \propto \frac{1}{E - E_n + i\Gamma/2}, \quad (21)$$

where E_n is the resonant energy of the motion along the normal to the film and Γ is the resonance width. The resonant energy E_n is determined from (18) and, thus, we have to determine the total resonance width Γ .

Near $\xi(E_n) = 2\pi n$, the exponential $\exp(-i\xi(E))$ can be written as $1 - i(E - E_n)d\xi/dE|_{E=E_n}$. Substituting this

expression into (20) and dividing the numerator and denominator by $-d\xi/dE|_{E=E_n}$, we find that

$$\begin{aligned} B(E_n) &= \frac{\vec{T}_1}{[1 - i(E - E_n)d\xi/dE|_{E=E_n} - \exp(-2k_i''L_i)|\vec{R}_1\vec{R}_2|]} \\ &= i \frac{dE}{d\xi|_{E=E_n}} \frac{\vec{T}_1}{E - E_n + i\Gamma/2}, \end{aligned} \quad (22)$$

where

$$\Gamma = \frac{1}{d\xi/dE|_{E=E_n}} (1 - \exp(-2k_i''L_i)|\vec{R}_1\vec{R}_2|). \quad (23)$$

The resonances are most pronounced at small widths, when $\vec{R}_1 \approx 1$ and $\vec{R}_2 \approx 1$. In this case, $\vec{R}_1 \approx 1 - |\vec{T}_1|^2/2$ and $\vec{R}_2 \approx 1 - |\vec{T}_2|^2/2$; therefore,

$$\Gamma = \Gamma_1 + \Gamma_2 + \Gamma_a. \quad (24)$$

Here,

$$\Gamma_1 = \frac{1}{d\xi/dE|_{E=E_n}} |\vec{T}_1|^2 \quad (25)$$

is the probability of emergence from the resonant layer through the left barrier,

$$\Gamma_2 = \frac{1}{d\xi/dE|_{E=E_n}} |\vec{T}_2|^2 \quad (26)$$

is the probability of emergence from the resonant layer through the right barrier, and

$$\begin{aligned} \Gamma_a &= \frac{2}{d\xi/dE|_{E=E_n}} (1 - \exp(-2k_i''L_i)) \\ &\approx \frac{4}{d\xi/dE|_{E=E_n}} k_i''L_i \end{aligned} \quad (27)$$

is the probability of absorption in the layer i . The latter approximate equality is valid when $2k_i''L_i \ll 1$.

When the main parameter, which determines the resonance, is $2k_i'(E_n)L_i \approx 2\pi n$, we have $d\xi/dE|_{E=E_n} \approx 2L_i/k_i'(E_n) \approx 2\pi n/k_i'^2(E_n)$. Therefore,

$$\begin{aligned} \Gamma_1 &= \frac{E_{ni}}{\pi n} |\vec{T}_1|^2, \quad \Gamma_2 = \frac{E_{ni}}{\pi n} |\vec{T}_2|^2, \\ \Gamma_a &= 2k_i'(E_n)k_i''(E_n), \end{aligned} \quad (28)$$

where $E_{ni} = k_i'^2(E_n)/2$ is the energy of motion of a neutron in the layer i along the normal to this layer. Thus, the total resonance width is the sum of three widths, Γ_1 , Γ_2 , and Γ_a , which are determined by the probabilities of

emergence from the resonant level through potentials 1 or 2 or the probability of absorption in the layer.

In this case, the squared modulus of the resonant factor (20) has the form

$$A(E) = |B(E)|^2 = \frac{\Gamma_1 E_{ni}/\pi n}{(E - E_n)^2 + \Gamma^2/4}; \quad (29)$$

i.e., it contains the input channel width in the numerator.

3. PROPERTIES OF THE REFLECTION AND TRANSMISSION AMPLITUDES OF POTENTIALS 1 AND 2

To determine the properties of the reflection amplitude R of the entire system, it is necessary to know the properties of all other amplitudes entering (8).

3.1. Properties of the Reflection and Transmission Amplitudes at $u_i = 0$

This case is interesting because it allows use of a special calculation technique, which will be applied below. At $u_i = 0$, the amplitudes have the following properties [27].

For any potential ($j = 1, 2$), the following relation is valid:

$$\vec{T}_{j0} = \vec{T}_{j0} = T_{j0} \quad (30)$$

(the additional subscript 0 indicates that $u_i = 0$). If the potentials are symmetric, the equalities

$$\vec{R}_{j0} = \vec{R}_{j0} = R_{j0} \quad (31)$$

are also valid. Let us use the representation

$$T_{j0} = |T_{j0}| \exp(i\varphi_{j0}) \quad (32)$$

with a real phase φ_{j0} . Then, in the absence of losses and in the case of symmetric potentials, we have

$$R_{j0} = \pm i |R_{j0}| \exp(i\varphi_{j0}) \quad (33)$$

with the same phase φ_{j0} . In view of the unitarity condition,

$$T_{j0}^2 - R_{j0}^2 = \exp(2i\varphi_{j0}). \quad (34)$$

At asymmetric potentials, the reflection amplitudes for the left and right sides are not equal but have equal magnitudes:

$$|\vec{R}_{j0}| = |\vec{R}_{j0}| \equiv |R_{j0}|. \quad (35)$$

Thus, instead of relation (34), we have

$$T_{j0}^2 - \vec{R}_{j0}\vec{R}_{j0} = \exp(2i\varphi_{j0}). \quad (36)$$

It follows from (35) that the reflection amplitudes for the two sides differ only in phases, and, from (36) and (33), we obtain

$$\vec{R}_{j0} \exp(-2i\varphi_{j0}) = -\vec{R}_{j0}^*, \quad \frac{\exp(2i\varphi_{j0})}{\vec{R}_{j0}} = -\frac{\vec{R}_{j0}}{|\vec{R}_{j0}|^2}, \quad (37)$$

where the asterisk denotes complex conjugation.

Losses lead to the appearance of imaginary parts in the potential and the phase.

3.2. Properties of Reflection and Transmission Amplitudes at $u_i \neq 0$

At $u_i \neq 0$, in order to reveal the properties of the reflection and transmission amplitudes, it is convenient to divide the entire system into parts with the use of infinitely narrow gaps, as shown in Fig. 2. This procedure makes it possible to express the amplitudes

$$\vec{R}_j, \quad \vec{T}_j \quad (38)$$

in terms of

$$\vec{R}_{j0}, \quad \vec{T}_{j0} \quad (39)$$

and write the reflection and transmission amplitudes of a rectangular potential step in the form

$$\begin{aligned} r_{0i} = -r_{oi} &= \frac{k_0 - k_i}{k_0 + k_i}, \quad t_{oi} = \frac{2k_0}{k_0 + k_i}, \\ t_{i0} &= \frac{2k_i}{k_0 + k_i} = \frac{k_i}{k_0} t_{oi}, \end{aligned} \quad (40)$$

where the left of the two subscripts in (40) shows on which side a neutron is incident. Taking into account the multiple rereflection of the neutron in the gaps between the potentials, we obtain [27]

$$\vec{R}_2 = r_{i0} + \frac{t_{i0} t_{oi} \vec{R}_{20}}{1 - r_{oi} \vec{R}_{20}} = -\frac{r_{oi} - \vec{R}_{20}}{1 - r_{oi} \vec{R}_{20}}, \quad (41)$$

$$\vec{R}_1 = r_{i0} + \frac{t_{i0} t_{oi} \vec{R}_{10}}{1 - \vec{R}_{10} r_{oi}} = -\frac{r_{oi} - \vec{R}_{10}}{1 - \vec{R}_{10} r_{oi}},$$

$$\vec{R}_1 = \vec{R}_{10} + \frac{T_{10} r_{oi}}{1 - \vec{R}_{10} r_{oi}} = \frac{\vec{R}_{10} + \exp(2i\varphi_{10}) r_{oi}}{1 - \vec{R}_{10} r_{oi}}, \quad (42)$$

$$\vec{T}_1 = \frac{T_{10} t_{oi}}{1 - \vec{R}_{10} r_{oi}} = \frac{k_0}{k_i} \vec{T}_{10}.$$

In (41), we took into account the equality $t_{oi} t_{io} - r_{oi} r_{io} = 1$. In (42), expression (36) and relation (40) between t_{oi} and t_{i0} were taken into consideration.

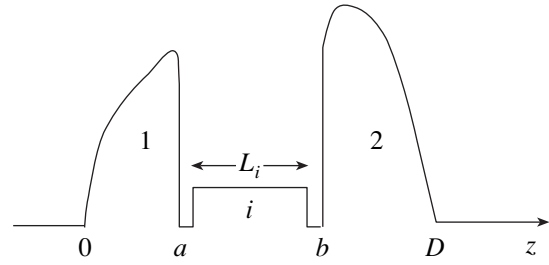


Fig. 2. The potential shown in Fig. 1 can be split into three parts by infinitely narrow gaps.

If $\vec{T}_j$ is written in the form

$$\vec{T}_j = |\vec{T}_j| \exp(i\varphi_j), \quad (43)$$

the factor $\vec{T}_1 \vec{T}_1 - \vec{R}_1 \vec{R}_1$ in the numerator of expression (8) is equal to $\exp(2i\varphi_1)$. Indeed, substituting expressions (41) and (42) into (8) and taking into account (37), we obtain

$$\begin{aligned} & \vec{T}_1 \vec{T}_1 - \vec{R}_1 \vec{R}_1 \\ &= \exp(2i\varphi_{10}) \frac{(1 - r_{oi}^2) |\vec{T}_{10}|^2 + (r_{oi} - \vec{R}_{10})(r_{oi} - \vec{R}_{10}^*)}{(1 - \vec{R}_{10} r_{oi})^2} \end{aligned} \quad (44)$$

$$= \exp(2i\varphi_{10}) \frac{1 - r_{oi} \vec{R}_{10}^*}{1 - r_{oi} \vec{R}_{10}} = \exp(2i\varphi_1),$$

where $\varphi_1 = \varphi_{10} - \arg(1 - r_{oi} \vec{R}_{10}^*)$. It is assumed that $k_i^0 > 0$; therefore, r_{oi} is a real number.

It follows from relations (16) and (42) that

$$|\vec{R}_1 \vec{R}_1| = |\vec{R}_1|^2 = |\vec{R}_1|^2 \equiv |R_1|^2 = 1 - |\vec{T}_1| |\vec{T}_1|. \quad (45)$$

Hence,

$$\frac{\exp(2i\varphi_1)}{\vec{R}_1} = -\frac{\vec{R}_1}{|R_1|^2}. \quad (46)$$

Expression (8) for $R(E)$, which depends on the energy E , is reduced to

$$R(E) = \frac{\vec{R}_1 |R_1|^2 - \vec{R}_1 \exp(2ik_i L_i) \vec{R}_2}{|R_1|^2 1 - \exp(2ik_i L_i) \vec{R}_1 \vec{R}_2}. \quad (47)$$

Under resonance conditions, expression (47) takes the form

$$R(E_n) = \frac{\vec{R} |R_1| - |R_2|}{|R_1| 1 - |R_1 R_2|}. \quad (48)$$

It follows from (48) that, at $|R_1| = |R_2|$, the reflectance $|R|^2$ becomes zero and, at $|R_2| = 1$ (i.e., in the case of

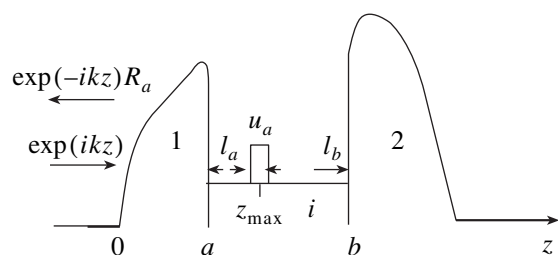


Fig. 3. A probe layer with a thickness Δ and potential u_a is located in the layer i shown in Fig. 1 at the point where the wave function amplitude is maximum.

total reflection from potential 2), the reflectance of the entire system is also equal to unity.

4. PROBING OF THE WAVE FUNCTION IN A LAYER

If a layer of an absorbent (for example, Gd) is placed at the point where the square of the wave function (10) is maximum, the absorption probability and the yield of nuclear reaction products must be maximum. The absorption probability for a layer with a potential $u_a = u'_a - iu''_a$ with respect to the layer i and a thickness Δ , located in the range $z_a \equiv z_{\max} - \Delta/2 \leq z \leq z_{\max} + \Delta/2 \equiv z_b$ in the layer i , as shown in Fig. 3, be can estimated from the quantity

$$W_0 = |\psi(z_{\max})|^2 N_0 \sigma_a \Delta = K u''_0 \frac{\Delta}{|k_i|}, \quad (49)$$

where σ_a is the absorption cross section of nuclei with the potential u_a . According to the optical theorem, σ_a is related to the imaginary part of the scattering amplitude by the expression $b'' = |k|\sigma/4\pi$. Therefore, $N_0 \sigma_a = u''_a/|k_i|$ and $|k_i|$ is the total wave vector rather than its normal component in the layer i . Under resonance conditions, this expression has the form

$$W_0 = \frac{(1 + |\vec{R}_2|^2) |\vec{T}_1|^2}{1 - |\vec{R}_1 \vec{R}_2|^2} u''_0 \frac{\Delta}{|k_i|}. \quad (50)$$

The absorption probability can also be calculated exactly. To this end, we will use the solution to the problem for the layer i and find the wave function in the layer a . The latter will be considered to be internal. The entire potential on the left of the point z_a will be denoted as 1a and the potential on the right of the point z_b will be denoted as 2a. The wave incident within the layer a on the edge z_b will be denoted as X_a . The reflection and transmission amplitudes of the entire system, including the layer with the potential u_a , will be denoted as R_a and T_a , respectively. Then, relations (7)–(10) will take the

form

$$X_a = \frac{\vec{T}_{1a} \exp(ik_a \Delta)}{1 - \exp(2ik_a \Delta) \vec{R}_{1a} \vec{R}_{2a}}, \quad (51)$$

$$R_a = \frac{\vec{R}_{1a} + (\vec{T}_{1a} \vec{T}_{1a} - \vec{R}_{1a} \vec{R}_{1a}) \exp(2ik_a \Delta) \vec{R}_{2a}}{1 - \exp(2ik_a \Delta) \vec{R}_{1a} \vec{R}_{2a}}, \quad (52)$$

$$T_a = \frac{\vec{T}_{2a} \vec{T}_{1a} \exp(ik_a \Delta)}{1 - \exp(2ik_a \Delta) \vec{R}_{1a} \vec{R}_{2a}}, \quad (53)$$

$$\psi_a(z) = [\exp(ik_a(z - z_b)) + \exp(-ik_a(z - z_b)) \vec{R}_{2a}] \times \frac{\exp(ik_a \Delta) \vec{T}_{1a}}{1 - \exp(2ik_a \Delta) \vec{R}_{1a} \vec{R}_{2a}}. \quad (54)$$

Here, the reflection and transmission amplitudes of potentials 1a and 2a have corresponding subscripts. They can easily be expressed in terms of the reflection and transmission amplitudes of potentials 1 and 2 and jumps of the potential a in the layer i . The corresponding expressions are written similarly to (41) and (42):

$$\begin{aligned} \vec{R}_{2a} &= r_{ai} + \frac{t_{ai} t_{ia} \exp(2ik_i l_b) \vec{R}_2}{1 - r_{ia} \exp(2ik_i l_b) \vec{R}_2} \\ &= \frac{r_{ai} + \exp(2ik_i l_b) \vec{R}_2}{1 - r_{ia} \exp(2ik_i l_b) \vec{R}_2}, \end{aligned} \quad (55)$$

where $l_b = b - z_b$ is the width of the part of the layer i between potentials 2 and u_a . Having denoted the width of the part of the layer i between the potentials 1 and u_a as $l_a = z_a - a$, we obtain, correspondingly,

$$\begin{aligned} \vec{R}_{1a} &= r_{ai} + \frac{t_{ai} t_{ia} \exp(2ik_i l_a) \vec{R}_1}{1 - r_{ia} \exp(2ik_i l_a) \vec{R}_1} \\ &= \frac{r_{ai} + \exp(2ik_i l_a) \vec{R}_1}{1 - r_{ia} \exp(2ik_i l_a) \vec{R}_1}, \end{aligned} \quad (56)$$

$$\begin{aligned} \vec{R}_{1a} &= \vec{R}_1 + \frac{\vec{T}_1 \vec{T}_1 \exp(2ik_i l_a) r_{ia}}{1 - r_{ia} \exp(2ik_i l_a) \vec{R}_1} \\ &= \frac{\vec{R}_1 + \exp(2ik_i l_a + 2i\phi_1) r_{ia}}{1 - r_{ia} \exp(2ik_i l_a) \vec{R}_1}, \end{aligned} \quad (57)$$

$$\vec{T}_{1a} = \frac{\vec{T}_1 \exp(ik_i l_a) t_{ia}}{1 - r_{ia} \exp(2ik_i l_a) \vec{R}_1} = \frac{k_a}{k} \vec{T}_{1a}, \quad (58)$$

$$\begin{aligned} r_{ia} &= -r_{ai} \frac{k_i - k_a}{k_i + k_a}, \quad t_{ai} = \frac{2k_a}{k_a + k_i}, \\ t_{ia} &= \frac{2k_i}{k_a + k_i} = \frac{k_i}{k_a} t_{ai}. \end{aligned} \quad (59)$$

The probability of absorption in the layer a is

$$W_a = \int_{-\Delta/2}^{\Delta/2} |\psi_a(z_{\max} + z)|^2 dz \frac{u_0''}{|k_a|}. \quad (60)$$

At small Δ , this expression is reduced to (50).

5. CALCULATION FOR SPECIFIC SYSTEMS

To calculate actual systems, it is necessary to specify potentials 1 and 2.

5.1. Zero Potentials 1 and 2

In this case, we assume that $a = 0$ and $b = L_i$. The reflection and transmission amplitudes are determined only by the potential jump at the points $z = 0$ and $z = L_i$:

$$\vec{R}_2 = \vec{R}_1 = r_{i0} = -r_{0i} = -\frac{k_0 - k_i}{k_0 + k_i}, \quad (61)$$

$$\vec{T}_1 = t_{0i} = \frac{2k_0}{k_0 + k_i}, \quad \vec{T}_2 = \vec{T}_1 = t_{i0} = \frac{2k_i}{k_0 + k_i}.$$

Substituting (61) into (10), we obtain

$$\psi_i = [\exp(ik_i(z - L_i)) + \exp(-ik_i(z - L_i))r_{0i}] \times \frac{\exp(ik_i L_i)t_{0i}}{1 - \exp(2ik_i L_i)r_{0i}^2}. \quad (62)$$

At small k_i , the transmission amplitude $t_{0i} \approx 2$ and the reflection amplitude r_{0i} is close to unity. Therefore, the gain of the wave function in the film can be significant [28]. Let us represent r_{0i} in the form $r_{0i} = 1 - t_{0i} \approx 1 - 2k_i/k_0$. Under the resonance conditions $2k_i L_i = 2\pi n$, the wave function (62) is approximated by the expression

$$\psi_i \approx i(-1)^n \sin(k_i(z - L_i)) \frac{k_0}{k_i} \quad (63)$$

and the gain is $K = (k_0/k_i)^2$.

In the presence of absorption in the film, the potential $u_i = u_i' - iu_i''$ and the wave vector $k_i = \sqrt{k_0^2 - u_i}$ are complex values. In this case, the gain at small k_i is

$$K = \left| \frac{4k_0}{4k_i' - [1 - \exp(-2k_i'' L_i)]k_0} \right|^2. \quad (64)$$

In weakly absorbing materials, at $k_i' = 0.1k_0 \approx 0.1\sqrt{u_i'}$, the imaginary part u_i'' of the potential can still be neglected, and the gain in resonance is $k_0^2/k_i'^2 = 100$. For example, in the case of bismuth, in which $2\pi/\sqrt{u_i'} = 1140 \text{ \AA}$ and $u_i''/u_i' = 10^{-6}$, in a resonant film of minimum thickness $L_i = 2\pi/k_i' = 1.14 \text{ \mu m}$, the value of

$2k_i'' L_i = k_i' L_i (u_i''/k_i'^2) = 10^{-4}\pi$ and can be neglected. Thus, in a 1.14- μm -thick bismuth film, the wave function density can be higher than the corresponding parameter in a vacuum by a factor of 100. According to (28), the width of the resonant state at $n = 1$ and in the absence of absorption is

$$\Gamma = \frac{4k_i'^3}{\pi k_0}. \quad (65)$$

At $k_i/k_0 = 0.1$, the level width is ~ 0.1 of the level height $k_i'^2$.

5.2. A Two-Layer System: A Layer on a Substrate

Let potential 1 be zero and potential 2 be a step of height u_2 of a semi-infinite substrate. In this case,

$$\vec{T}_1 = t_{0i} = \frac{2k_0}{k_0 + k_i}, \quad \vec{R}_1 = r_{i0} = -r_{0i} = -\frac{k_0 - k_i}{k_0 + k_i}, \quad (66)$$

$$\vec{R}_2 = r_{i2} = \frac{k_i - k_2}{k_i + k_2}, \quad k_2 = \sqrt{k_0^2 - u_2}. \quad (67)$$

Substituting (66) and (67) into (10), we obtain

$$\psi_i(z) = [\exp(ik_i z) + \exp(-ik_i z) \exp(2ik_i L_i) r_{i2}] \times \frac{t_{0i}}{1 + \exp(2ik_i L_i) r_{0i} r_{i2}}. \quad (68)$$

At the point $z = 0$, the wave function has the form

$$\psi_i(0) = [1 + \exp(2ik_i L_i) r_{i2}] \frac{t_{0i}}{1 + \exp(2ik_i L_i) r_{0i} r_{i2}} = 1 + R_{i2}. \quad (69)$$

Here, the latter equality follows from the continuity of the function at the film boundary and R_{i2} denotes the reflection amplitude for the entire substrate–film system. This amplitude has the form

$$R_{i2} = R_{i0} + \frac{T_{i0}^2 r_{02}}{1 - R_{i0} r_{02}} = \frac{R_{i0} + \exp(2i\phi_{i0}) r_{02}}{1 - R_{i0} r_{02}}, \quad (70)$$

where R_{i0} and T_{i0} are the reflection and transmission amplitudes of the film i located in a vacuum:

$$\begin{aligned} R_{i0} &= r_{0i} \frac{1 - \exp(2ik_i L_i)}{1 - r_{0i}^2 \exp(2ik_i L_i)}, \\ T_{i0} &= e^{ik_i L_i} \frac{1 - r_{0i}^2}{1 - r_{0i}^2 \exp(2ik_i L_i)}, \\ \exp(2i\phi_{i0}) &= \frac{\exp(2ik_i L_i) - r_{0i}^2}{1 - r_{0i}^2 \exp(2ik_i L_i)}. \end{aligned} \quad (71)$$

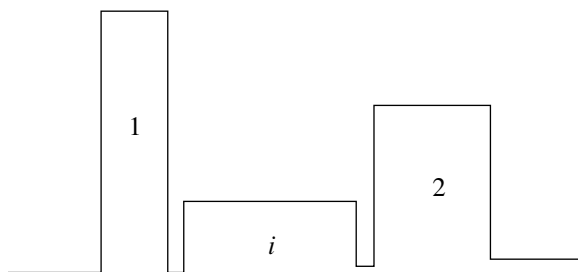


Fig. 4. A system composed of three films. For calculations, it is convenient to introduce infinitely narrow gaps between the films.

5.3. A Thin Absorbent Layer on a Film Located on a Substrate

If the value of $1 + R_{i2}$ at the film surface is known, the probability of absorption in a thin absorbing layer deposited on its surface can be estimated from the quantity

$$W = N_{0a} \sigma_a L_a |1 + R_{i2}|^2, \quad (72)$$

where N_{0a} is the atomic density, σ_a is the scattering cross section of a nucleus, and L_a is the film thickness. The yield of a nuclear reaction can be calculated exactly by two methods. The first method implies calculation of the amplitudes of reflection (R) and transmission (T) of the three-layer system (absorbent u_a with a thickness L_a , the film u_i with a thickness L_i , and a substrate u_2):

$$R = R_{a0} + T_{a0}^2 \frac{R_{i2}}{1 - R_{a0} R_{i2}} = \frac{R_{a0} + \exp(2i\phi_{a0}) R_{i2}}{1 - R_{a0} R_{i2}}, \quad (73)$$

where

$$\begin{aligned} R_{a0} &= r_{0a} \frac{1 - \exp(2ik_a L_a)}{1 - \exp(2ik_a L_a) r_{0a}^2}, \\ T_{a0} &= \exp(ik_a L_a) \frac{1 - r_{0a}^2}{1 - \exp(2ik_a L_a) r_{0a}^2}, \\ \exp(2i\phi_{a0}) &= \frac{\exp(2ik_a L_a) - r_{0a}^2}{1 - \exp(2ik_a L_a) r_{0a}^2}. \end{aligned} \quad (74)$$

Then, the difference $W = 1 - |R|^2 - |T|^2 k_2 / k_0$ is found. At small film thicknesses, we should obtain the same result in the approximation that is linear in L_a (72).

The second method consists in calculation of the exact wave function $\psi_a(z)$ in the absorbing layer:

$$\begin{aligned} &\psi_a(z) \\ &= \lfloor \exp(ik_a(z - L_a)) + \exp(-ik_a(z - L_a)) \vec{R}_{ai2} \rfloor X_a, \\ &X_a = \frac{\exp(ik_a L_a) t_{0a}}{1 + \exp(2ik_a L_a) \vec{R}_{ai2} r_{0a}}, \end{aligned} \quad (75)$$

where

$$\begin{aligned} \vec{R}_{ai2} &= \frac{-r_{0a} + R_{i2}}{1 + r_{02} R_{i2}}, \quad r_{0a} = \frac{k_0 - k_a}{k_0 + k_a}, \\ t_{0a} &= \frac{2k_0}{k_0 + k_a}, \quad k_a = \sqrt{k_0^2 - u_a} = k'_a + ik''_a, \end{aligned} \quad (76)$$

with subsequent calculation of the quantity

$$W_a = \int_0^{L_a} |\psi_a(z)|^2 dz N_0 \sigma. \quad (77)$$

We omit the result of integration in (77) in view of its awkwardness.

5.4. Resonances in a Three-Layer System

Let us assume that potentials 1 and 2 are formed by two films with widths $L_{1,2}$ and optical potentials $u_{1,2}$; i.e., the entire multilayer system consists of three films. To determine all desired reflection and transmission amplitudes, it is convenient to represent a system consisting of three films as three rectangular potentials separated by infinitely narrow gaps (Fig. 4) and use the results obtained in Subsection 2.2.

In the case of three films, we can use expressions (41) and (42) and substitute, instead of R_{j0} and T_{j0} , symmetric reflection and transmission amplitudes for individual rectangular potentials of height u_j and width L_j into these expressions:

$$\begin{aligned} R_{j0} &= r_{0j} \frac{1 - \exp(2ik_j L_j)}{1 - r_{0j}^2 \exp(2ik_j L_j)}, \\ T_{j0} &= \exp(ik_j L_j) \frac{1 - r_{0j}^2}{1 - r_{0j}^2 \exp(2ik_j L_j)}, \end{aligned} \quad (78)$$

where

$$r_{0j} = \frac{k_0 - k_j}{k_0 + k_j}, \quad k_j = \sqrt{k_0^2 - u_j}. \quad (79)$$

As a result, we obtain

$$\begin{aligned} \vec{R}_2 &= r_{i0} + \frac{t_{i0} t_{0i} R_{20}}{1 - r_{0i} R_{20}} = -\frac{r_{0i} - R_{20}}{1 - r_{0i} R_{20}}, \\ \vec{R}_1 &= r_{i0} + \frac{t_{i0} t_{0i} R_{10}}{1 - R_{10} r_{0i}} = -\frac{r_{0i} - R_{10}}{1 - R_{10} r_{0i}}, \end{aligned} \quad (80)$$

$$\begin{aligned} \vec{R}_1 &= R_{10} + \frac{T_{10}^2 r_{0i}}{1 - R_{10} r_{0i}} = \frac{R_{10} + \exp(2i\phi_{10}) r_{0i}}{1 - R_{10} r_{0i}}, \\ \vec{T}_1 &= \frac{T_{10} t_{0i}}{1 - R_{10} r_{0i}} = \frac{k_0}{k_i} \vec{T}_1. \end{aligned} \quad (81)$$

If $u_i \ll k_0^2 \leq u_{1,2}$ and $L_{1,2}$ are sufficiently large,

$$\vec{T}_1 \approx \exp(-|k_1|L_1)t_{0i} = \frac{k_0}{k_i}\vec{T}_1 \quad (82)$$

and

$$\begin{aligned} |\vec{R}_{1,2}| &\approx 1 - \frac{1}{2}|\vec{T}_{1,2}\vec{T}_{1,2}| \\ &= 1 - 2\exp(-2|k_{1,2}|L_{1,2})\frac{k_0|k_{1,2}|}{|u_{1,2}|}. \end{aligned}$$

Therefore, the neutron density gain in a material under resonance conditions and at $|k_1|L_1 < |k_2|L_2$ can reach the value

$$\begin{aligned} K &= \frac{|\vec{T}_1|^2(1+|\vec{R}_2|)^2}{(1-|\vec{R}_1||\vec{R}_2|)^2} \\ &= \frac{4u_1}{|k_1|^2} \frac{\exp(2|k_1|L_1)}{1 + 2|u_1k_2/u_2k_1|\exp(2[|k_1|L_1 - |k_2|L_2])}. \end{aligned} \quad (83)$$

If $|k_2|L_2 \gg |k_1|L_1$, the gain is

$$K = \frac{4u_1}{|k_1|^2} \exp(2|k_1|L_1). \quad (84)$$

In this case, the widths of the resonant levels are $\Gamma_i \propto \exp(-2|k_i|L_i)$. These expressions were derived with disregard for the imaginary parts of the potentials. Consideration of the imaginary parts will decrease the gain. At specified real and imaginary parts of the potentials, one can calculate the maximum value of the gain corresponding to certain layer thicknesses. This problem does not meet fundamental difficulties, but we will not dwell on it.

5.5. Potentials 1 and 2 in the Form of Periodic Chains

Let potentials 1 and 2 be periodic chains with $N_{1,2}$ bilayers in each chain [29–31]. For simplicity, we will assume that a chain consists of identical symmetric periods. One of such periods is shown in Fig. 5. It is formed by a barrier of height u_b and width l_b and two wells at both sides of it. The well potential and the well width are u_w and $l_w/2$, respectively. Thus, the period width is $a = l_b + l_w$.

The reflection amplitude R_∞ of a symmetric periodic potential with an infinite number of periods is [32]

$$R_\infty = \frac{\sqrt{(1+r)^2 - t^2} - \sqrt{(1-r)^2 - t^2}}{\sqrt{(1+r)^2 - t^2} + \sqrt{(1-r)^2 - t^2}}, \quad (85)$$

where r and t is the reflection and transmission amplitudes of one period. If the phases of two square roots

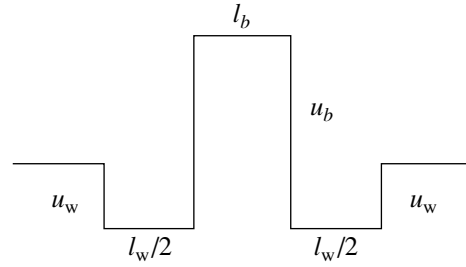


Fig. 5. The symmetrized period of the periodic potential, which is composed of a rectangular potential barrier with the height u_b and width l_b and a potential well with the depth u_w and width l_w .

differ by $\pi/2$, the reflection amplitude becomes a complex unity, which will be represented as $\exp(-2i\phi)$. In this case, total Bragg reflection occurs. The total reflection range and the width of this range are referred to as the Darwin table and the width of the Darwin table, respectively.

The reflection and transmission amplitudes of N periods are written similarly to (78):

$$\begin{aligned} R_N &= R_\infty \frac{1 - \exp(2iqNa)}{1 - R_\infty^2 \exp(2iqNa)}, \\ T_N &= \exp(iqNa) \frac{1 - R_\infty^2}{1 - R_\infty^2 \exp(2iqNa)}, \end{aligned} \quad (86)$$

where q is the Bloch wave vector, which is determined from the expression

$$\exp(iqa) = \frac{\sqrt{(1+t)^2 - r^2} - \sqrt{(1-t)^2 - r^2}}{\sqrt{(1+t)^2 - r^2} + \sqrt{(1-t)^2 - r^2}}. \quad (87)$$

When the energy of motion of neutrons in the direction that is perpendicular to the layers lies within the total reflection range, i.e., within the Darwin table, the Bloch wave vector contains an imaginary part iq'' and the amplitude R_N is completely analogous to the reflection amplitude for a rectangular potential barrier, for which the incident neutron energy is below the barrier height. In this case, as for a three-layer system, the neutron density gain in the layer i under resonance conditions can be estimated from the quantity

$$K \propto \exp(2q''N_1a) \quad (88)$$

if $N_2 \gg N_1$ and

$$K \propto \exp(2q''N_2a) \quad (89)$$

if $N_1 \gg N_2$. The maximum value of q'' can be estimated from the value $q'' \leq u/k_B$, where u is the mean of the period potential and $k_B = \pi n/a$ is the Bragg wave number.

5.6. A Magnetic Layer with a Magnetization That Is Noncollinear with the External Magnetic Field

Let us consider a situation where potentials 1 and 2 are absent; i.e., we have a single layer i . Let this layer be magnetic and located in an external magnetic field; the layer magnetization is noncollinear with the external magnetic field. In this case, the wave function in the layer is determined from expression (6), in which the amplitudes $\hat{R}_2$, $\hat{R}_1$, and $\hat{T}_1$ are determined by the jumps of the potential and the field at the layer interfaces [33, 34]:

$$\hat{R}_1 = -\hat{R}_2 = -\hat{R}_2 = \hat{r}_{0i} = -\hat{r}_{i0} = (\hat{k} + \hat{k}_i)^{-1}(\hat{k} - \hat{k}_i), \quad (90)$$

$$\hat{T}_1 = \hat{t}_{0i} = 2(\hat{k} + \hat{k}_i)^{-1}\hat{k} = \hat{I} + \hat{r}_{0i}, \quad (91)$$

$$\hat{T}_1 = \hat{t}_{i0} = 2(\hat{k} + \hat{k}_i)^{-1}\hat{k}_i = \hat{I} - \hat{r}_{0i},$$

where $\hat{k} = \sqrt{k_0^2 - 2\sigma\mathbf{B}_0}$, $\hat{k}_i = \sqrt{k_0^2 - u_i - 2\sigma\mathbf{B}_i}$, and k_0 is the wave vector in a vacuum in the absence of a field.

To determine the spin-flip and non-spin-flip reflection amplitude, we will use expression (4) and substitute the amplitudes from (91) into it:

$$\begin{aligned} \hat{R} &= \hat{R}_i = \hat{r}_{0i} \\ &- \hat{t}_{i0} \exp(i\hat{k}_i L_i) \hat{r}_{0i} [\hat{I} - \exp(i\hat{k}_i L_i) \hat{r}_{0i} \exp(i\hat{k}_i L_i) \hat{r}_{0i}]^{-1} \\ &\times \exp(i\hat{k}_i L_i) \hat{t}_{0i}. \end{aligned} \quad (92)$$

At small L_i , in the zero-order approximation in powers of L_i , we have

$$\begin{aligned} \hat{R}_i^{(0)} &= \hat{r}_{0i} - \hat{t}_{i0} \hat{r}_{0i} [\hat{I} - \hat{r}_{0i}^2]^{-1} \hat{t}_{0i} \\ &= \hat{r}_{0i} - [\hat{I} - \hat{r}_{0i}^2]^{-1} \hat{t}_{i0} \hat{t}_{0i} \hat{r}_{0i} = 0. \end{aligned} \quad (93)$$

This result is quite natural because, at $L_i = 0$, the layer is absent as well as the reflection from it. In the next (linear) order, we obtain

$$\begin{aligned} i \frac{\hat{R}_i^{(1)}}{L_i} &= \hat{t}_{i0} \hat{k}_i \frac{\hat{r}_{0i}}{\hat{I} - \hat{r}_{0i}^2} \hat{t}_{0i} + \hat{t}_{i0} \frac{\hat{r}_{0i}}{\hat{I} - \hat{r}_{0i}^2} \hat{k}_i \hat{t}_{0i} \\ &+ \hat{t}_{i0} \frac{\hat{r}_{0i}}{\hat{I} - \hat{r}_{0i}^2} (\hat{k}_i \hat{r}_{0i}^2 + \hat{r}_{0i} \hat{k}_i \hat{r}_{0i}) \frac{1}{\hat{I} - \hat{r}_{0i}^2} \hat{t}_{0i} \\ &= \frac{1}{\hat{I} + \hat{r}_{0i}} \hat{k}_i \frac{\hat{r}_{0i}}{\hat{I} - \hat{r}_{0i}} + \frac{\hat{r}_{0i}}{\hat{I} + \hat{r}_{0i}} \hat{k}_i \frac{1}{\hat{I} - \hat{r}_{0i}} = \frac{1}{2\hat{k}} [\hat{k}^2 - \hat{k}_i^2]. \end{aligned} \quad (94)$$

Since $\hat{k}_i^2 = k_0^2 - u_i - 2\mathbf{B}_i\sigma$ and we are interested only in the matrices leading to a spin flip, expression (94) can be simplified as

$$\Delta \hat{R}_i = \frac{-iL_i}{\hat{k}} \mathbf{B}_\perp \sigma,$$

where $\mathbf{B}_\perp = \mathbf{B}_i - \mathbf{B}(\mathbf{B}\mathbf{B}_i)/B^2$.

5.7. A Magnetic Layer in a Nonmagnetic System

Let the layer i be located between nonmagnetic potentials 1 and 2 and the entire system be placed in a uniform external field B that is noncollinear with the internal field $\mathbf{B}_i$ of the layer i . Splitting the system, as shown in Fig. 2, we can write the total reflection amplitude as

$$\hat{R} = \hat{R}_{10} + \hat{T}_{10} \hat{R}_{i2} [\hat{I} - \hat{R}_{10} \hat{R}_{i2}]^{-1} \hat{T}_{10}, \quad (95)$$

where $\hat{R}_{i2}$ is the matrix amplitude of the total reflection of the layer i and potential 2:

$$\hat{R}_{i2} = \hat{R}_i + \hat{T}_i \hat{R}_{20} [\hat{I} - \hat{R}_i \hat{R}_{20}]^{-1} \hat{T}_i. \quad (96)$$

Here, $\hat{R}_i$ and $\hat{T}_i$ are the reflection and transmission amplitudes of the layer i located in a vacuum and exposed to a magnetic field B ,

$$\begin{aligned} \hat{R}_i &= \hat{r}_{i0} - \hat{t}_{0i} \exp(i\hat{k}_i L_i) \hat{r}_{0i} \\ &\times [\hat{I} - \exp(i\hat{k}_i L_i) \hat{r}_{0i} \exp(i\hat{k}_i L_i) \hat{r}_{0i}]^{-1} \exp(i\hat{k}_i L_i) \hat{t}_{0i}, \end{aligned} \quad (97)$$

$$\begin{aligned} \hat{T}_i &= \hat{t}_{i0} [\hat{I} - \exp(i\hat{k}_i L_i) \hat{r}_{0i} \exp(i\hat{k}_i L_i) \hat{r}_{0i}]^{-1} \\ &\times \exp(i\hat{k}_i L_i) \hat{t}_{0i}. \end{aligned} \quad (98)$$

At small L_i , in the linear approximation in L_i , we have

$$\begin{aligned} \hat{R}_i &\approx \frac{-iL_i}{2\hat{k}} (\hat{k}^2 - \hat{k}_i^2) \approx \frac{-iL_i}{\hat{k}} \mathbf{B}_\perp \sigma, \\ \hat{T}_i &\approx \hat{I} + \frac{iL_i}{2\hat{k}} (\hat{k}^2 + \hat{k}_i^2) \approx \hat{I} - \frac{iL_i}{\hat{k}} \mathbf{B}_\perp \sigma. \end{aligned} \quad (99)$$

In equalities (97) and (98), we left only those terms proportional to L_i that lead to a spin flip. Therefore, in the linear approximation, we have

$$\begin{aligned} \hat{R}_{i2} &\approx \hat{R}_{20} - \frac{iL_i}{2\hat{k}} (\hat{k}^2 - \hat{k}_i^2) + \frac{iL_i}{2\hat{k}} (\hat{k}^2 + \hat{k}_i^2) \hat{R}_{20} \\ &+ \hat{R}_{20} \frac{iL_i}{\hat{k}} \hat{k}_i^2. \end{aligned} \quad (100)$$

We are interested only in the spin-flip related part of expression (100). This part is given by

$$\begin{aligned} \hat{R}_{i2} &= \hat{R}_{20} + \Delta \hat{R}_i \\ &\approx \hat{R}_{20} - \frac{iL_i}{\hat{k}} [\mathbf{B}_\perp \sigma + \mathbf{B}_\perp \sigma \hat{R}_{20} + \hat{R}_{20} \mathbf{B}_\perp \sigma]. \end{aligned} \quad (101)$$

A linear correction to the total reflection amplitude has

the form

$$\Delta \hat{R} = \hat{T}_{10} [\Delta \hat{R}_{i2} + \hat{R}_{20} [\hat{I} - \hat{R}_{10} \hat{R} \hat{R}_{20}]^{-1} \hat{R}_{10} \Delta \hat{R}_{i2}] \times [\hat{I} - \hat{R}_{10} \hat{R}_{20}]^{-1} \hat{T}_{10}. \quad (102)$$

Substituting $\Delta \hat{R}_{i2}$ from (101) into (102), we obtain

$$\Delta \hat{R} = -i \frac{\hat{T}_{10}}{\hat{I} - \hat{R}_{10} \hat{R}_{20}} \frac{L_i}{2\hat{k}} \times [\mathbf{B}_\perp \boldsymbol{\sigma} + \mathbf{B}_\perp \hat{\boldsymbol{\sigma}} \hat{R}_{20} + \hat{R}_{20} \mathbf{B}_\perp \boldsymbol{\sigma}] \frac{\hat{T}_{10}}{\hat{I} - \hat{R}_{10} \hat{R}_{20}}. \quad (103)$$

Expression (103) takes into account that diagonal matrices commute with each other. It follows from (103) that the spin-flip reflection amplitude is proportional to the squared amplitude of the wave function in the region where the magnetic layer is located. This proportionality is evidenced by the two factors

$\hat{T}_{10}/[\hat{I} - \hat{R}_{10} \hat{R}_{20}]$, each of which determines the wave function amplitude in this region. Therefore, the spin-flip reflection amplitude is proportional to the gain of the wave function, which can be high in resonance and compensates the smallness of the parameter $k_i L_i$.

In [35, 36], the reflectances and absorptances were calculated for a multilayer system Be(h)/Bi(19 nm)/Fe(1.45 Å, $M_i = 0.1 \mu_B$, $B = 5$ Oe, $\alpha(B, B_i) = 10^\circ$)/Bi(31 nm)/Be(400 nm)/Si, where α is the angle between the external field B and the internal field B_i . With an increase in the thickness h of the first layer from zero to 50 nm, the spin-flip reflectance increases by five orders of magnitude and the absorptance increases by three orders of magnitude.

5.8. Resonances in a Thick Magnetic Film

Let us consider a magnetic layer in the regions having potentials 1 and 2 when the entire system is located in an external field that is noncollinear with the layer magnetization. Amplification of the wave function in the layer is determined from the matrix in (3) and mainly from the inverse matrix

$$[\hat{I} - \exp(i\hat{k}_i L_i) \hat{R}_1 \exp(i\hat{k}_i L_i) \hat{R}_2]^{-1}. \quad (104)$$

To determine the gain, it is necessary to find the eigenstates and eigenvalues of the matrix

$$\hat{Y} = \exp(i\hat{k}_i L_i) \hat{R}_1 \exp(i\hat{k}_i L_i) \hat{R}_2. \quad (105)$$

Let us assume the quantization axis to be directed along the vector $\mathbf{B}_i$ and write $\hat{k}_i$ in the form $k(\boldsymbol{\sigma}_z B_i)$. If the external field is the same on both sides of the magnetic

film, $\hat{R}_1$ and $\hat{R}_2$ can be written as

$$\begin{aligned} \hat{R}_1 &= r_0^{(1)} + r_z^{(1)} \boldsymbol{\sigma}_z + r_x^{(1)} \boldsymbol{\sigma}_x, \\ \hat{R}_2 &= r_0^{(2)} + r_z^{(2)} \boldsymbol{\sigma}_z + r_x^{(2)} \boldsymbol{\sigma}_x. \end{aligned} \quad (106)$$

As a result, we can transform the matrix $\hat{Y}$ as

$$\begin{aligned} \hat{Y} &= \exp(2ik(\boldsymbol{\sigma}_z B_i) L_i) (r_0^{(1)} + r_z^{(1)} \boldsymbol{\sigma}_z) \hat{R}_2 \\ &+ \exp(i[k(B_i) + k(-B_i)] L_i) \\ &\times [(r_0^{(2)} - r_z^{(2)} \boldsymbol{\sigma}_z) r_x^{(1)} \boldsymbol{\sigma}_x + r_x^{(1)} r_x^{(2)}]. \end{aligned} \quad (107)$$

The diagonal and off-diagonal elements of the matrix $\hat{Y}$ can easily be found from (107):

$$\begin{aligned} Y_{uu} &= \exp(2ik(B_i) L_i) (r_0^{(1)} + r_z^{(1)}) (r_0^{(2)} + r_z^{(2)}) \\ &+ \exp(i[k(B_i) + k(-B_i)] L_i) r_x^{(1)} r_x^{(2)}, \end{aligned} \quad (108)$$

$$\begin{aligned} Y_{dd} &= \exp(2ik(-B_i) L_i) (r_0^{(1)} - r_z^{(1)}) (r_0^{(2)} - r_z^{(2)}) \\ &+ \exp(i[k(B_i) + k(-B_i)] L_i) r_x^{(1)} r_x^{(2)}, \end{aligned} \quad (109)$$

$$\begin{aligned} Y_{ud} &= \exp(2ik(B_i) L_i) (r_0^{(1)} + r_z^{(1)}) \\ &+ \exp(i[k(B_i) + k(-B_i)] L_i) (r_0^{(2)} - r_z^{(2)}) r_x^{(1)}, \end{aligned} \quad (110)$$

$$\begin{aligned} Y_{du} &= \exp(2ik(-B_i) L_i) (r_0^{(1)} - r_z^{(1)}) \\ &+ \exp(i[k(B_i) + k(-B_i)] L_i) (r_0^{(2)} + r_z^{(2)}) r_x^{(1)}, \end{aligned} \quad (111)$$

where $k(\pm B_i) = \sqrt{k_0^2 - u_i \mp B_i}$.

Using these matrix elements, one can easily find the eigenvalues Λ_i of the matrix $\hat{Y}$ by solving the equation

$$\det(\hat{Y} - \hat{\Lambda}) = 0. \quad (112)$$

It follows from (112) that

$$\Lambda_{1,2} = \frac{1}{2} [Y_{uu} + Y_{dd} \pm \sqrt{(Y_{uu} - Y_{dd})^2 - 4Y_{ud}Y_{du}}]. \quad (113)$$

With known values of Λ_i , we find the eigenspinors $|\xi_{1,2}^\Lambda\rangle$ by solving the equation

$$\hat{Y} |\xi_{1,2}^\Lambda\rangle = \Lambda_{1,2} |\xi_{1,2}^\Lambda\rangle. \quad (114)$$

The matrix $\hat{Y}$ can always be written as $\hat{Y} = Y_0 + \mathbf{Q}\boldsymbol{\sigma}$ with some vector $\mathbf{Q}$. Then, $\hat{Y} = Y_0 + \mathbf{Q}\boldsymbol{\sigma}$ and $\Lambda_{1,2} = Y_0 \pm Q$. Note that $Y_0 = (Y_{uu} + Y_{dd})/2$, $Q_z = (Y_{uu} - Y_{dd})/2$, $Q_x = (Y_{du} + Y_{ud})/2$, $Q_y = (Y_{du} - Y_{ud})/2i$, and $Q = \sqrt{Q_x^2 + Q_y^2 + Q_z^2}$.

The normalized eigenstates can be written as

$$\begin{aligned} |\xi_{1,2}^\Lambda\rangle &= \frac{Q \pm Q\sigma}{\sqrt{2Q(Q \pm Q_z)}} |\xi_u^i\rangle \\ &= \frac{\sqrt{Q \pm Q_z}}{\sqrt{2Q}} |\xi_u^i\rangle \pm \frac{Q_x + iQ_y}{\sqrt{2Q(Q \pm Q_z)}} |\xi_d^i\rangle, \end{aligned} \quad (115)$$

where $|\xi_{u,d}^i\rangle$ denote the eigenstates polarized along the internal field and in the opposite direction, respectively. Expression (115) is valid at $Q_x \neq 0$ and (or) $Q_y \neq 0$.

The normalized eigenstate $|\xi\rangle = \alpha|\xi_u^i\rangle + \beta|\xi_d^i\rangle$ corresponds to the spin arrow $\mathbf{S} = \langle \xi | \boldsymbol{\sigma} | \xi \rangle = (S_x, S_y, S_z)$ with the component S_z equal to $|\alpha|^2 - |\beta|^2$. Therefore, the eigenstates (115) are polarized at angles $\theta_{1,2}$ to the internal field; the cosines of these angles are

$$\begin{aligned} \cos\theta_{1,2} &= \frac{|Q \pm Q_z|^2 - |Q_x + iQ_y|^2}{2|Q||Q \pm Q_z|} \\ &= \frac{|Q_z|^2 - \text{Im}(Q_x Q_y^*) \pm \text{Re}(Q^* Q_z)}{|Q||Q \pm Q_z|}. \end{aligned}$$

With the use of $|\xi_{1,2}^\Lambda\rangle$, expressions (3)–(6) can be written as:

$$\begin{aligned} \hat{X} &= [\hat{I} - \exp(i\hat{k}_i L_i) \hat{R}_1 \exp(i\hat{k}_i L_i) \hat{R}_2]^{-1} \exp(i\hat{k}_i L_i) \hat{T}_1 \\ &= \sum_{j=1,2} |\xi_j^\Lambda\rangle \frac{1}{1 - \Lambda_j} \langle \xi_j^\Lambda | \exp(i\hat{k}_i L_i) \hat{T}_1, \end{aligned} \quad (116)$$

$$\begin{aligned} \hat{R} &= \hat{R}_1 + \hat{T}_1 \exp(i\hat{k}_i L_i) \hat{R}_2 \\ &\times [\hat{I} - \exp(i\hat{k}_i L_i) \hat{R}_1 \exp(i\hat{k}_i L_i) \hat{R}_2]^{-1} \exp(i\hat{k}_i L_i) \hat{T}_1 \\ &= \hat{R}_1 + \sum_{j=1,2} \hat{T}_1 \exp(i\hat{k}_i L_i) \hat{R}_2 |\xi_j^\Lambda\rangle \frac{1}{1 - \Lambda_j} \langle \xi_j^\Lambda | \\ &\times \exp(i\hat{k}_i L_i) \hat{T}_1, \end{aligned} \quad (117)$$

$$\hat{T} = \hat{T}_2 \hat{X} = \sum_{j=1,2} \hat{T}_2 |\xi_j^\Lambda\rangle \frac{1}{1 - \Lambda_j} \langle \xi_j^\Lambda | \exp(i\hat{k}_i L_i) \hat{T}_1, \quad (118)$$

$$\begin{aligned} |\psi_i\rangle &= [\exp(i\hat{k}_i(z-b)) + \exp(-i\hat{k}_i(z-b)) \hat{R}_2] \hat{X} |\xi_0\rangle \\ &= [\exp(i\hat{k}_i(z-b)) + \exp(-i\hat{k}_i(z-b)) \hat{R}_2] \\ &\times \sum_{j=1,2} |\xi_j^\Lambda\rangle \frac{1}{1 - \Lambda_j} \langle \xi_j^\Lambda | \exp(i\hat{k}_i L_i) \hat{T}_1 |\xi_0\rangle. \end{aligned} \quad (119)$$

It can be seen that amplification occurs when Λ_j is close to unity under the condition that the resonant states in

the magnetic film are polarized at some angle to the internal field.

Let us consider the case of a single magnetic film placed in a vacuum, where the matrix reflection and transmission amplitudes are given by expressions (90) and (91). We will find the spin-flip and non-spin-flip reflection amplitudes $R^{\pm,\pm}$ for such a film and analyze the influence of resonances in the film on these amplitudes.

The eigenstates $|\xi_\pm^0\rangle$ with polarization directed along the external field $\mathbf{B}_0$ and in the opposite direction can be written as

$$|\xi_\pm^0\rangle = \frac{1}{\sqrt{2B_0(B_0 \pm B_{0z})}} (B_0 \pm \mathbf{B}_0 \boldsymbol{\sigma}) |\xi_u^i\rangle, \quad (120)$$

where the z axis is assumed to be directed along the internal field. Accordingly, the elements of the reflection matrix can be written as

$$\begin{aligned} R^{\pm,\pm} &= \frac{1}{2B_0(B_0 \pm B_{0z})} \langle \xi_u^i | \left[(B_0 \pm \mathbf{B}_0 \boldsymbol{\sigma}) \left(\hat{r}_{0i} \right. \right. \\ &+ \sum_{j=1,2} [1 - \hat{r}_{i0}] \exp(i\hat{k}_i L_i) \hat{r}_{i0} |\xi_j^\Lambda\rangle \\ &\times \frac{1}{1 - \Lambda_j} \langle \xi_j^\Lambda | \exp(i\hat{k}_i L_i) [1 + \hat{r}_{0i}] \left. \right) (B_0 \pm \mathbf{B}_0 \boldsymbol{\sigma}) \left. \right] |\xi_u^i\rangle. \end{aligned} \quad (121)$$

Let a resonance occur in the film at some eigenvalue Λ_j . Then, a matrix element can be approximated by the resonant term

$$\begin{aligned} R^{\pm,\pm} &\approx \frac{1}{2B_0(B_0 \pm B_{0z})} \\ &\times \langle \xi_u^i | \left[(B_0 \pm \mathbf{B}_0 \boldsymbol{\sigma}) \left([1 - \hat{r}_{i0}] \exp(i\hat{k}_i L_i) \hat{r}_{i0} |\xi_1^\Lambda\rangle \right. \right. \\ &\times \frac{1}{1 - \Lambda_1} \langle \xi_1^\Lambda | \exp(i\hat{k}_i L_i) [1 + \hat{r}_{0i}] \left. \right) (B_0 \pm \mathbf{B}_0 \boldsymbol{\sigma}) \left. \right] |\xi_u^i\rangle. \end{aligned} \quad (122)$$

Thus, to find the spin-flip and non-spin-flip reflection amplitudes, we have to calculate the matrix elements

$$\begin{aligned} &\langle \xi_u^i | (B_0 \pm \mathbf{B}_0 \boldsymbol{\sigma}) [1 - \hat{r}_{i0}] \exp(i\hat{k}_i L_i) \hat{r}_{i0} |\xi_1^\Lambda\rangle, \\ &\langle \xi_1^\Lambda | \exp(i\hat{k}_i L_i) [1 + \hat{r}_{0i}] (B_0 \pm \mathbf{B}_0 \boldsymbol{\sigma}) |\xi_u^i\rangle. \end{aligned} \quad (123)$$

Note that, in the case of a film, the matrix $\hat{Y}$ in (90) is equal to the squared matrix $\hat{y} = \exp(i\hat{k}_i L_i) \hat{r}_{0i}$; therefore, $\Lambda_j = \lambda_j^2$, where λ_j are the eigenvalues of the matrix $\exp(i\hat{k}_i L_i) \hat{r}_{0i}$:

$$\exp(i\hat{k}_i L_i) \hat{r}_{0i} |\xi_j^\Lambda\rangle = \lambda_j |\xi_j^\Lambda\rangle,$$

where $|\xi_j^\Lambda\rangle$ are the eigenvectors of this matrix.

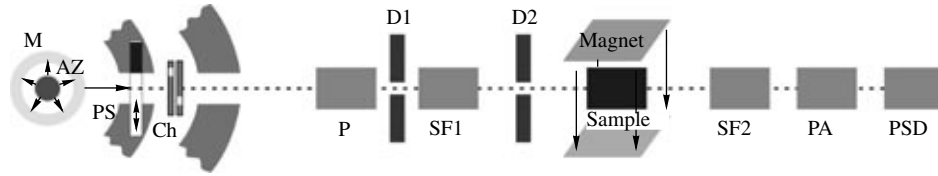


Fig. 6. Block diagram of the REMUR spectrometer: (AZ) reactor active zone, (M) water moderator, (Ch) neutron beam chopper, (PS) protective shutter, (P) polarizer, (D1 and D2) cadmium diaphragms, (SF1 and SF2) spin flippers, (PA) polarization analyzer, and (PSD) position-sensitive detector.

Since the matrix $\hat{r}$ can be written as $\hat{r} = r_0 + r_z\sigma_z + r_x\sigma_x$, the elements of the matrix $\hat{y}$ have the form

$$\begin{aligned} y_{uu} &= \exp(ik(B_i)L_i)(r_0 + r_z), \\ y_{dd} &= \exp(ik(-B_i)L_i)(r_0 - r_z), \end{aligned} \quad (124)$$

$$y_{ud} = \exp(ik(B_i)L_i)r_x, \quad y_{du} = \exp(ik(-B_i)L_i)r_x.$$

The eigenvalues λ_j are

$$\lambda_{1,2} = \frac{1}{2}[y_{uu} + y_{dd} \pm \sqrt{(y_{uu} - y_{dd})^2 - 4y_{ud}y_{du}}]. \quad (125)$$

When the eigenvalues λ_i are known, one can easily find the eigenspinors $|\xi_{1,2}^\lambda\rangle$ by solving the equation

$$\hat{y}|\xi_{1,2}^\lambda\rangle = \lambda_{1,2}|\xi_{1,2}^\lambda\rangle. \quad (126)$$

The matrix $\hat{y}$ can always be written as $\hat{y} = y_0 + \mathbf{q}\sigma$ with some vector $\mathbf{q}$. Then, $y_{1,2} = y_0 \pm q$ and the normalized eigenstates have the form

$$|\xi_{1,2}^\lambda\rangle = \frac{q \pm \mathbf{q}\sigma}{\sqrt{2q(q \pm q_z)}}|\xi_u^i\rangle. \quad (127)$$

The desired matrix elements (122) are now represented as

$$\begin{aligned} R^{\pm,\pm} &\approx \frac{-1}{2B_0\sqrt{(B_0 \pm B_{0z})} \cdot \sqrt{(B_0 \pm B_{0z})}} \\ &\times \langle \xi_u^i | \left[(B_0 \pm \mathbf{B}_0\sigma) \left((1 - \hat{r}_{0i})|\xi_1^\lambda\rangle \right. \right. \\ &\times \left. \left. \frac{\lambda_1}{1 - \lambda_1^2} \langle \xi_1^\lambda | \exp(ik(B_i)L_i)(1 + \hat{r}_{0i}) \right) (B_0 \pm \mathbf{B}_0\sigma) \right] | \xi_u^i \rangle. \end{aligned} \quad (128)$$

Thus, to find the spin-flip and non-spin-flip reflection amplitudes, one has to calculate the matrix elements

$$\begin{aligned} &\langle \xi_u^i | (B_0 \pm \mathbf{B}_0\sigma) [1 - \hat{r}_{0i}] | \xi_1^\lambda \rangle, \\ &\langle \xi_1^\lambda | \exp(ik(B_i)L_i) [1 + \hat{r}_{0i}] (B_0 \pm \mathbf{B}_0\sigma) | \xi_u^i \rangle. \end{aligned} \quad (129)$$

We do not give the final formulas because they are extremely cumbersome. However, when necessary, they can be obtained and used to fit experimental data. Generally, in the case of arbitrary magnetic fields and

magnetizations of layers, one can use the numerical generalized matrix method to calculate the reflectances and wave functions in layers [20].

6. EXPERIMENTAL INVESTIGATIONS

In this section, we will review some experiments with multilayer systems carried out on the REMUR reflectometer (Fig. 6) of the IBR-2 reactor at the Joint Institute for Nuclear Research (Dubna). The wave function was probed in layered media by measuring the reflectances and transmittances of mirrors and detecting secondary radiations. Recently, the spectrometer has been equipped with a wide-spectral wide-aperture polarization instrument, which made it possible to increase the spectrometer's possibilities significantly.

6.1. REMUR Polarized Neutron Spectrometer

A detailed description of the design and the parameters of the REMUR spectrometer can be found in [37]. Here, we will only note its specific features that are directly related to study of neutron standing waves.

The spectrometer is located at channel 8 of the IBR-2 reactor, where the neutron flux in a water moderator into a solid angle of 2π sr is $\Phi_0 = 4 \times 10^{12} \text{ cm}^{-2} \text{ s}^{-1}$. At the point where a sample is located, the flux density of polarized neutrons with a vertical divergence of $2\theta_0 = 2.5 \text{ mrad}$ and the same horizontal divergence is $\Phi_s = 2 \times 10^6 \text{ cm}^{-2} \text{ s}^{-1}$. The spectrum of neutrons has a Maxwellian shape with a temperature of 300 K. The operating neutron wavelength range of the spectrometer is 1–10 Å. With a repetition frequency of power pulses of the IBR-2 reactor of 5 Hz, measurements are performed within successive time intervals of 200 ms.

The wavelength of neutrons detected is determined from the time of flight through the distance from the source (water moderator located near the active zone of the reactor) to the detector. At a flight distance of 34 m, the mean-square deviation of the neutron wavelength, which is related to the finite pulse width of thermal neutrons in the moderator ($\tau = 0.32 \text{ ms}$), is $1.9 \times 10^{-2} \text{ Å}$ in the entire operating wavelength range.

The spectrometer facilities for polarizing neutrons and analyzing their polarization make it possible to investigate magnetic layered structures. The spectrometer contains a neutron polarizer, a system of dia-

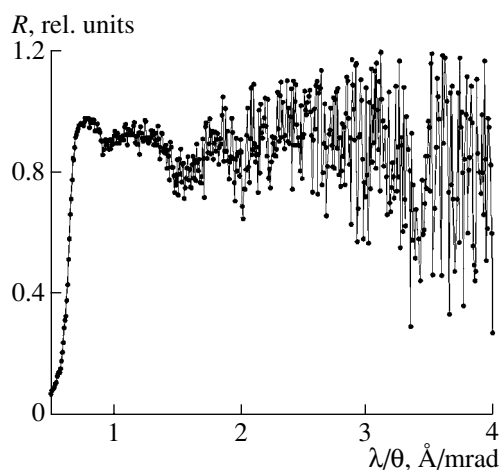


Fig. 7. Experimental dependence of the neutron reflectance $R(\lambda/\theta)$ for a Cu(30 nm)/Ti(150 nm)/Cu(100 nm)/glass structure.

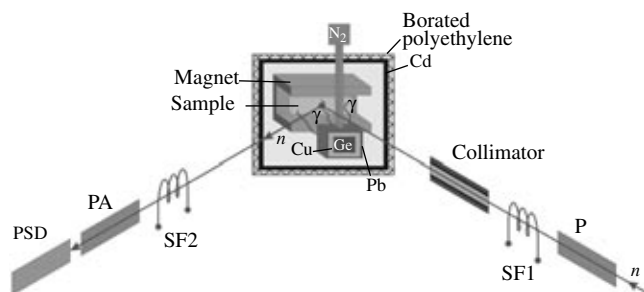


Fig. 8. Schematic of the γ -ray detection channel in the REMUR spectrometer.

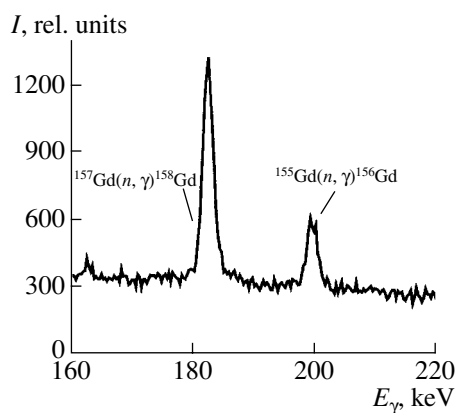


Fig. 9. Experimental spectrum of γ photons emitted from the gadolinium layer upon reflection of neutrons from a Gd(5 nm)/Fe(100 nm)/glass(5 mm) structure.

phragms, two spin flippers (before the sample and after it), a focused (fan-shaped) polarization analyzer, and a position-sensitive detector. The two spin flippers are used to measure the four reflectances of neutrons, R^{++} , R^{+-} , R^{-+} , and R^{--} , where the left and right superscripts

correspond to incident and scattered neutrons, respectively; the superscript + corresponds to the state with a spin oriented along the external magnetic field; and the superscript – corresponds to the state with an oppositely oriented spin. The reflectances R^{+-} and R^{-+} characterize the spin-flip reflection, while R^{--} and R^{++} characterize the non-spin-flip reflection.

6.2. Possibilities of the REMUR Spectrometer

The minimum resonance widths Γ that can be measured on the REMUR spectrometer are determined from the relation $\epsilon_k = \delta k/k_0 = \Gamma/k_0^2$, where ϵ_k is the relative resolution in k . At a wavelength of $\lambda = 1.5$ Å, this resolution is $\epsilon_k = \delta k/k_0 = \delta\lambda/\lambda \approx 1 \times 10^{-2}$. If we assume that the gain of the wave function is $K \approx k_0^2/\Gamma$, the maximum gain in a film that can be observed at a given value of $\lambda = 1.5$ Å is $K \approx 100$. Let us try to find out how the measurement of such a gain is statistically guaranteed. Under the assumption that the angular resolutions are $2\theta_y = 2.5 \times 10^{-3}$ and $2\delta\theta_x = 2\delta\theta_x \epsilon_{\theta_x} = 2.5 \times 10^{-5}$, $\epsilon_\lambda = 10^{-2}$, the beam cross-sectional area on the sample is $S = 0.1$ cm², and the measurement time is 1 h, we find that, for example, at a reflectance of $R = 10^{-3}$, the total number of reflected polarized neutrons is

$$\Delta N = \frac{\Phi_0}{2\pi} 2\theta_y 2\theta_x 2\epsilon_\lambda S R \times 3.6 \times 10^3 / 2 = 1.5 \times 10^2.$$

Thus, a measurement time of an hour ensures a statistical error of about 10%.

6.3. Probing of the Wave Function by the Specific Features on the Reflection Curve

Resonances in layers of multilayer mirrors lead to the occurrence of specific features on the reflection and transmission curves. Figure 7 shows the dependence of the reflectance $R\left(\frac{\lambda}{\theta}\right)$ for a Cu(30 nm)/Ti(150 nm)/

Cu(100 nm)/glass structure measured in [23] with resolutions of $\delta\lambda/\lambda \approx 2.6 \times 10^{-3}$ and $\delta\theta/\theta \approx 1.5 \times 10^{-3}$. It can be seen that, at $\lambda/\theta = 3.5$ Å/mrad, the reflectance R decreases to 0.6. This value corresponds to the absorption coefficient $\mu = 0.4$. Taking into account the imaginary part of the potential of titanium and the velocity of a neutron of 2.6 m/s in a Ti film, we find that, at a double passage of a 150-nm-thick film, the probability of neutron absorption is 5.8×10^{-3} . Obviously, the total absorption probability can be as high as $\mu = 0.4$ if the film-averaged gain is 70 ($K = 140$).

6.4. Probing of the Wave Function by the Yield of Nuclear Reactions Related to Neutron Trapping

Resonances in layers can be additionally probed using the nuclear reaction yield or the specific features of the change in polarization under reflection. Let us consider the technique of probing of wave functions by the yield of nuclear reactions that occur upon trapping of a neutron in a resonant layer. Neutron trapping leads to nuclear reactions of four possible types: (n, γ) , (n, p) , (n, α) , and $(n, \text{particles} + \text{fission fragments})$. The reaction with escape of γ photons (n, γ) is most widespread. The cross sections of interaction of thermal neutrons with nuclei at energies of 10^{-3} to 10^{-1} eV range from 10^{-3} to 10^5 barn.

6.4.1. Probing of the wave function at the mirror surface by the (n, γ) reaction yield. A layered Gd(5 nm)/Fe(100 nm)/glass structure was prepared to probe the wave function at the mirror surface [38]. An iron film deposited on a glass substrate served as a mirror. The wave function density at its surface was measured from the yield of γ photons upon neutron trapping in the thin gadolinium layer deposited on the iron film.

Figure 8 shows a schematic of the setup based on beam 8 of the IBR-2 reactor [39]. An important element of the system is the small-aperture input collimator, which serves both to collimate the thermal neutron flux and to decrease the background counts of the γ detector. Such a collimator makes it possible to decrease the distance between the γ detector and a sample to 25 cm. A germanium detector equipped with an anti-Compton shield for suppressing the background by a factor of 3–5 was used as a detector of γ photons with energies of several hundred kiloelectronvolts. To decrease the background count level, the detector was enclosed in a passive combined protective coating composed of a 5-cm-thick borated polyethylene layer, a 10-cm-thick lead layer, and a 1-cm-thick copper layer (external, intermediate, and internal protective layers, respectively).

Figure 9 shows the measured spectra of γ photons emitted from the layer natural gadolinium nuclei. Despite the measures undertaken, the background count level remained significant. In this context, polarized neutrons were used to increase the reliability of the standing-wave probing.

The ratio $\eta(\lambda)$ of the yield of γ photons for neutrons polarized along the field to the yield of γ photons for neutrons with the opposite polarization was experimentally measured (Fig. 10). The difference of the ratio $\eta(\lambda)$ from unity is related to the difference in the heights of the potential barrier in the iron for the two spin states. For neutrons polarized along the magnetic field, the potential barrier is $U^+ = 341$ neV, while, for the neutrons with the opposite polarization, $U^- = 80.5$ neV. This difference in the barrier heights leads to the experimentally observed characteristic long-wavelength dependence $\eta(\lambda)$ [39].

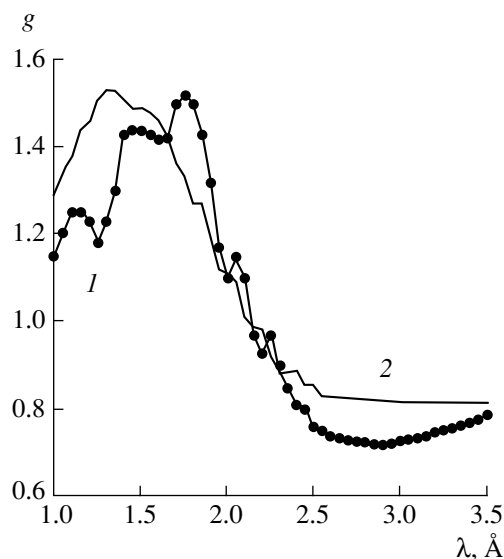


Fig. 10. (1) Experimental and (2) calculated dependences of the polarization ratio g of the γ -photon count on the neutron wavelength.

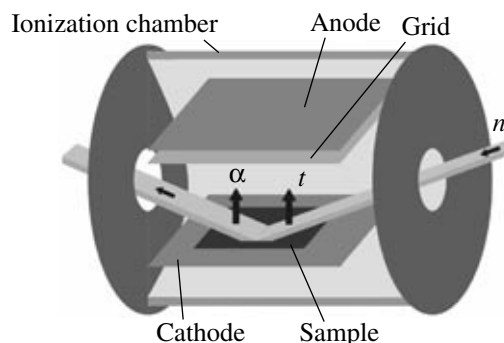


Fig. 11. Ionization chamber with the sample studied ($^6\text{LiF}(20 \text{ nm})/\text{Ti}(200 \text{ nm})/\text{Cu}(100 \text{ nm})/\text{glass}$), which was used to detect α particles and tritons emitted by a layer of nuclei of the ^6Li isotope after trapping the neutrons reflected from the structure.

As was expected, the ratio $\eta(\lambda)$ turned out to be larger than unity at $E_{\perp} > U^-$ and smaller than unity at $E_{\perp} < U^-$. However, the calculated curve insufficiently describes the experimental data. For example, the maximum in the experimental curve is shifted to longer wavelengths, while the experimental minimum value is smaller than the calculated one. Apparently, the incomplete agreement between the calculated and experimental data is explained by the insufficient magnetization of the iron layer in an external magnetic field of 300 Oe. In addition, it is not inconceivable that the magnetization vector is rotated at the boundary between the iron and gadolinium layers. This rotation should lead to a shift of the wavelengths λ_{max} and λ_{min} (the experimental values are 1.9 Å and 2.8 Å, respectively), at which the maximum and minimum values of η were observed

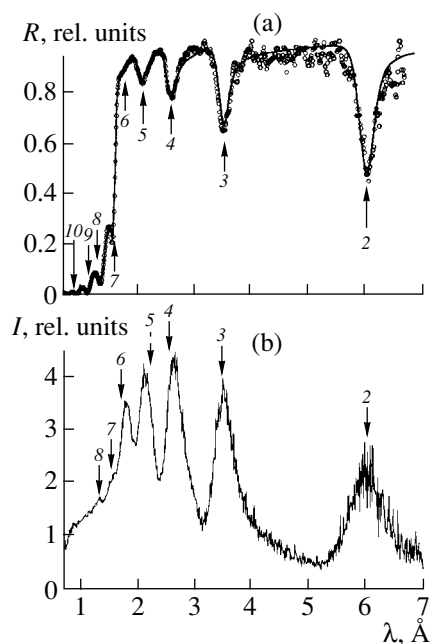


Fig. 12. Long-wavelength dependences of the (a) neutron reflectance for a ${}^6\text{LiF}(20\text{ nm})/\text{Ti}(200\text{ nm})/\text{Cu}(100\text{ nm})/\text{glass}$ structure and (b) yield of charged particles as a result of trapping of neutrons by ${}^6\text{Li}$ isotope nuclei. The resonance numbers are indicated.

(the calculated values are $\lambda_{\text{max}} = 1.4\text{ \AA}$ and $\lambda_{\text{min}} = 2.6\text{ \AA}$), as well as to a change in the value of η .

6.4.2. Probing of the wave function by the charged-particle yield. The reaction of neutron trapping by a nucleus with emission of charged particles is less widespread in comparison with the reaction of neutron trapping with emission of γ photons. For example, at cross sections $\sigma > 0.1$ barn, only six isotopes (${}^3\text{He}$, ${}^7\text{Be}$, ${}^{14}\text{N}$, ${}^{35}\text{Cl}$, ${}^{40}\text{K}$, and ${}^{76}\text{Br}$) are necessary for the (n, p) reaction, while for the (n, α) reaction, the number of necessary isotopes increases to 14 (${}^6\text{Li}$, ${}^{10}\text{B}$, ${}^{14}\text{N}$, ${}^{17}\text{O}$, ${}^{21}\text{Ne}$, ${}^{33}\text{S}$, ${}^{40}\text{K}$, ${}^{59}\text{Ni}$, ${}^{143}\text{Nd}$, ${}^{149}\text{Sm}$, ${}^{152}\text{Gd}$, ${}^{153}\text{Gd}$, ${}^{162}\text{Er}$, and ${}^{203}\text{Tl}$). For a fission reaction with emission of neutrons and charged fragments, the number of isotopes is large. For example, at relatively large reaction cross sections ($\sigma(n, f) > 30$ barn), such a reaction involves 18 isotopes (the corresponding elements are Ra, Th, U, Np, Pu, Am, Cm, and Cf). The conditions for detecting charged particles and γ photons are significantly different as well because a charged particle ionizes a medium. Hence, the charged-particle range is small (several tenths of a millimeter in air and several micrometers in a solid). This circumstance determines the design of an experimental setup, in which a sample under study is placed inside the charged-particle detector. As a result, the solid angle of measurement of charged particles is 2π sr. Owing to the small charged-particle range, the background charged-particle flux does not reach the detector, a fact that determines the low level of background count.

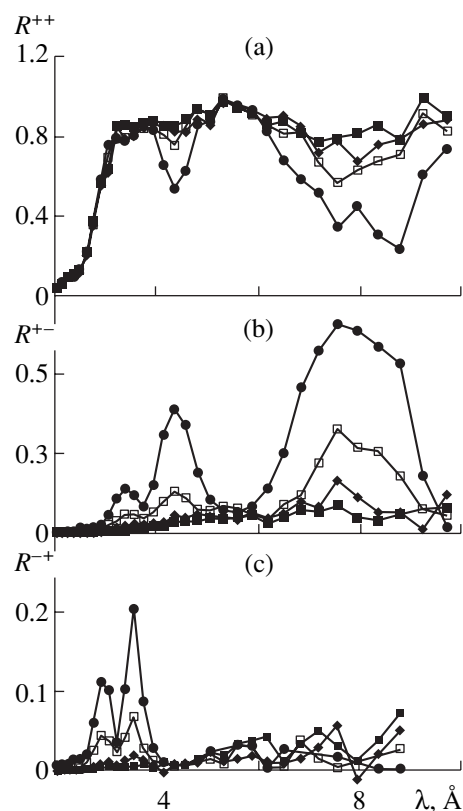


Fig. 13. Long-wavelength dependence of the neutron reflectances for a $\text{Cu}(10\text{ nm})/\text{Ti}(50\text{ nm})/\text{Fe}(12\text{ nm})/\text{Ti}(150\text{ nm})/\text{Cu}(100\text{ nm})/\text{glass}(5\text{ mm})$ structure at an external magnetic field of 4.5 kOe directed at the angles $\beta =$ (■) 0°, (◆) 10°, (□) 25°, and (●) 80° to the sample plane: (a) R^{++} , (b) R^{+-} , and (c) R^{-+} .

Figure 11 shows a schematic of the experiment [17] in which α particles and tritons were detected in an ionization chamber [40]. The sample (initially, a ${}^6\text{LiF}(20\text{ nm})/\text{Ti}(200\text{ nm})/\text{Cu}(100\text{ nm})/\text{glass}(5\text{ mm})$ structure) was located in the ionization chamber in such a way that the copper and titanium layers played the role of a cathode.

Figure 12 shows the long-wavelength count dependences for the neutrons reflected from the structure and charged particles emitted as a result of the neutron trapping by ${}^6\text{Li}$ isotope nuclei. These curves have a resonant character and correlate with each other. The correlation consists in the fact that a decrease in the count rate of reflected neutrons corresponds to the increase in the count rate of charged particles. This circumstance confirms that the minima in the neutron reflection are related to their absorption, while the emission of α particles and tritons is the result of the neutron absorption. Obviously, the resonant character of the curves is associated with the increase in the neutron density at “resonant” wavelengths. Fitting of the calculation results to the experimental data shows that, in the third resonance ($\lambda = 6\text{ \AA}$), the increase in the neutron density with respect to the density in the incident beam is $K = 35$.

The reason for this increase is that the ^6LiF layer not only absorbs but also reflects neutrons, in particular, owing to the imaginary part of the optical interaction potential.

6.5. Probing of the Wave Function Density by the Spin-Flip Reflection of Polarized Neutrons

The neutron wave function can be probed as well using a thin magnetic layer whose magnetization is noncollinear with an external field. Such a layer leads to spin flip during reflection and transmission. The intensity of the spin-flip reflection or transmission can be used to estimate the neutron density at the points of location of the magnetic layer when the angle between the magnetization vector of the layer and the external field is known.

Figure 13 shows the wavelength dependences of the reflectances R^{++} , R^{+-} , and R^{-+} for neutrons reflected from the Cu(10 nm)/Ti(50 nm)/Fe(12 nm)/Ti(150 nm)/Cu(100 nm)/glass(5 mm) structure at different angles (0° , 10° , 25° , and 80°) between the external magnetic field $B_0 = 4.5$ kOe and the sample plane [41]. It can be seen that the dependences of R^{++} and R^{+-} anticorrelate with each other, as the dependences in Fig. 12, which correspond to the detection of beams of reflected neutrons and α particles. The positions of maxima and minima correspond to the spatial position of the iron layer in the titanium layer.

6.5.1. Method for detecting spin-flip processes by neutron beam splitting. If the external field is sufficiently strong, spin-flip reflection leads to a deviation of the neutron propagation direction from that corresponding to the specular reflection [19, 33, 34, 41, 42]. When the incident beam is unpolarized, the reflected beam splits into three beams.

One of these beams propagates at the mirror angle and contains neutrons that were reflected without polarization rotation. The beam with the angle of reflection smaller than the angle of incidence contains neutrons that are completely polarized along the field. This beam is formed as a result of the spin flip of the neutrons that were polarized oppositely to the external field direction in the incident beam. The beam with the angle of reflection exceeding the angle of incidence contains neutrons that are completely polarized against the field. This beam is formed as a result of the spin flip of the neutrons that were polarized along the field in the incident beam.

Thus, measuring the intensities of coherent nonspecularly reflected components (they must be separated from the neutrons undergoing incoherent or small-angle scattering by inhomogeneities of the system or by interface irregularities), one can find the probability of spin-flip reflection without analyzing the polarization and thus determine the wave function density in the region of magnetic layer location if the angle between the layer magnetization and the external field

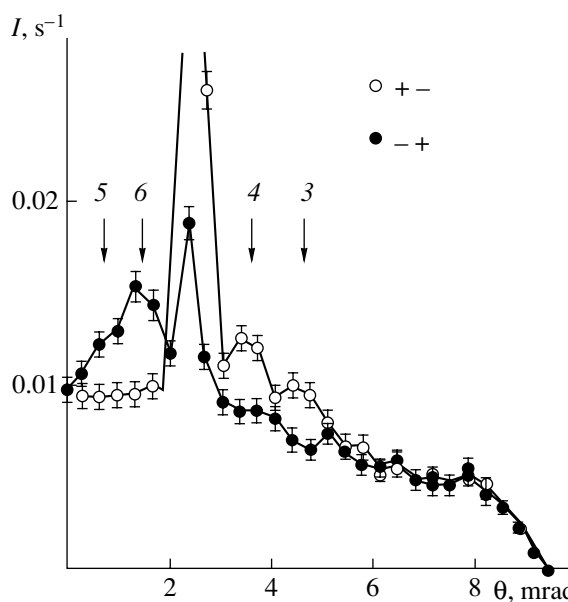


Fig. 14. Dependence of the intensity of spin-flip neutron reflection on the grazing angle for a Ti(30 nm)/Co(6 nm)/Ti(200 nm)/Cu(100 nm)/glass structure. The resonance numbers are indicated.

is known. Figure 14 shows the grazing-angle dependences of the reflectances R^{+-} and R^{-+} for the neutrons that are spin-flip reflected from the Ti(30 nm)/Co(6 nm)/Ti(200 nm)/Cu(100 nm)/glass(5 mm) system. It can be seen that $R^{+-} > R^{-+}$ for the grazing angle θ exceeding the specular angle $\theta_0 = 2.5$ mrad and, vice versa, $R^{+-} < R^{-+}$ at $\theta < \theta_0$. These relations are a manifestation of the splitting effect. At the same time, it can be seen that the reflectance R^{+-} has maxima at grazing angles $\theta_0 = 3.5$ and 4.5 mrad. These maxima, depending on the grazing angle of the incident beam, correspond at certain wavelengths to the standing-wave maxima in the cobalt layer for the spin state $+$.

It is noteworthy that the beam splitting effect can also be seen in Fig. 13. It manifests itself in the fact that, at $\lambda > 4$ Å, the spin-flip reflectance R^{+-} becomes zero. The reason for this phenomenon is that the energy of neutron motion in the direction normal to the film surface becomes negative at spin flip; i.e., such a neutron cannot escape from the film and passes to the state of a surface wave, which can propagate only along the film. In this case, the splitting is so strong that the angle of deviation of the spin-flip beam exceeds the grazing angle of the incident beam.

7. CONCLUSIONS

A specific mode of a neutron wave field—standing waves generated in a layered structure owing to the interfaces—has been considered. Standing waves arise above any part of a layered structure that strongly reflects neutrons. A particular mode—amplified stand-

ing waves that exist in an individual layer of the structure upon reflection of neutrons from both layer boundaries—has been described. Methods for calculating magnetic and nonmagnetic structures have been developed and structures of different type have been considered. In particular, it has been shown that the neutron wave amplitude at the vacuum/medium interface can increase by a factor of 2 in some cases, while the absorption in a thin layer between the film and substrate may increase by a factor of 4. For a three-layer system, it has been shown that, when neutron absorption is disregarded, the gain of neutron density increases unlimitedly with increasing thickness of the boundary layers.

An important situation where the boundary layers are periodic structures themselves has been considered. In this case, the wavelength of a standing wave can be as small as an interatomic distance, whereas, upon total neutron reflection from continuous layers, it is several hundred angstroms. It has been shown that the gain exponentially increases with an increase in the number of bilayers and the ratio of squares of the critical and Bragg wave vectors.

The case of a magnetic layer located between nonmagnetic layers has been considered. It has been shown that there is an increase in the spin-flip reflectance of neutrons, which is due to the resonances in which the spin in the layer makes some angle with the internal field.

The results of the experimental investigation of the standing wave field have been reported. Experiments have been described in which the secondary radiations related to the neutron trapping by nuclei in the standing-wave mode were detected. It has been shown that, at a value of the wave vector that is close to critical, a standing wave antinode is formed on the iron layer surface for the spin state of a neutron that corresponds to the initial polarization of the incident neutron beam. This secondary radiation is typical of most nuclei and can be used to characterize them in investigations. However, to reduce the background count level significantly, it is necessary to design a measuring system very thoroughly.

The results of the first experiment, where charged particles (α particles and tritons) served as secondary radiation, have been reported. Here, the gain was 35. This value was obtained using the reflection of neutrons from a strongly absorbing LiF layer. Note that the number of nuclei emitting α particles and protons is relatively small; however, there are many transuranium nuclei that decompose into fragments. With allowance for the fact that measurement of charged particles is accompanied by a low background count level, this secondary radiation can effectively be used in investigations.

It is known that polarized neutrons make it possible to investigate magnetic layered structures. Standing neutron waves with a specified spin state of neutrons allow determination of the position of a magnetic non-

collinear layer and, via the spin flip enhancement, increase the sensitivity of determining the degree of magnetic noncollinearity. These possibilities of polarized neutrons have been demonstrated. Spin-flipped neutrons have been detected by both direct measuring of the reflected beam polarization and recording of the angular neutron distribution (the effect of beam splitting in a magnetic field).

Obviously, the above problems do not cover the entire range of the opening experimental possibilities. No doubt, the mode of amplified standing waves has wide potential applications. With a further development of the technology of fabrication of perfect nanostructures, a practical possibility of implementing the mode of amplified standing waves with nanometer and even angstrom wavelengths will arise [6].

Some fundamental questions, such as the coherent length of a neutron and the description of its wave function by a wave packet, remained beyond the scope of this study; they were considered in [43, 44].

ACKNOWLEDGMENTS

We are grateful to M.V. Kovalchuk, who initiated to a great extent the investigations of neutron standing waves and took active participation in the discussions on the subject, and to N.A. Gundorin, F. Rada, Yu.N. Gledenov, P.V. Sedyshev, S.I. Zheludeva, V.A. Somenkov, L. Cher, S.V. Kozhevnikov, A.V. Petrenko, and V.V. Proglyado for their help.

This study was supported by the Rosnauka state contracts nos. 02.445.11.7003 and 02.434.11.2033, grant no. INTAS 03-51-6426 and 03-51-4778 and the Russian Foundation for Basic Research (project no. 04-02-16052).

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Translated by Yu. Sin'kov